MODULAR KOSZUL DUALITY FOR CHIRAL ALGEBRAS

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Abstract. The product of two chiral fields on an algebraic curve diverges where the fields collide. Among differential forms with a pole along the diagonal, exactly one has a residue independent of every auxiliary choice: the logarithmic form $\eta = d\log(z_1 - z_2)$. We show that η generates a differential on the Fulton–MacPherson compactification of configuration space, that the vanishing of its square is equivalent to the Borcherds identity, and that the resulting bar construction closes, over the moduli space of curves, into a single Maurer–Cartan element $\Theta_{\mathcal{A}}$ in a modular convolution Lie algebra. The equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ holds identically: it is the algebraic form of $\partial^2 = 0$ on the Deligne–Mumford boundary, and it holds identically, term by term. Every invariant constructed in this paper is a projection of $\Theta_{\mathcal{A}}$: the modular characteristic κ at degree two, the classical r-matrix as the binary collision residue, an obstruction tower whose generating function satisfies an algebraic equation of degree two, and a genus expansion whose scalar part is κ times the $\widehat{A}$-genus. The ordered form of the construction remembers strictly more than its symmetric quotient; the discrepancy is a quantum group. At positive genus the construction acquires a curvature, κ times the first Chern class of the Hodge bundle, and this single scalar is simultaneously the conformal anomaly of two-dimensional physics and the coefficient of the Hodge classes at every genus.

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1. Introduction

1.1. **The seed.** Let X be a smooth algebraic curve over $\mathbb{C}$ and let $a(z)$, $b(w)$ be chiral fields on X. Their product diverges as $z \to w$. Any regularisation of this divergence requires a 1-form on the configuration space of two points with a pole along the diagonal, and among such forms exactly one has a residue that does not depend on a coordinate, a metric, or a level:

$$\eta_{12} \;=\; d\log(z_1 - z_2) \;=\; \frac{dz_1 - dz_2}{z_1 - z_2}. \tag{1.1}$$

Three facts about η generate everything in this paper.

First, consistency at three points forces a relation. On $\mathrm{Conf}_3(\mathbb{C})$,

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0, \tag{1.2}$$

which is the partial-fractions identity in disguise; Arnold proved that this one relation generates the full ideal of relations in $H^*(\mathrm{Conf}_n(\mathbb{C}))$ [Ar]. Second, extraction of residues against η converts the operator product expansion into a differential, and the vanishing of the square of that differential is exactly the associativity axiom of a chiral algebra. Third, η fails to be invariant under a change of coordinate $z \mapsto w(z)$ by the cocycle $d\log w'(z)$; this failure is the central extension that the Virasoro algebra measures. One 1-form, one relation, one extraction: everything that follows is forced.

1.2. **The tower.** The objects of the theory arrange themselves in a tower, and the paper climbs it:

$$\begin{array}{ccccc} \text{factorization category} & \longrightarrow & \mathcal{A} = \mathrm{End}(b) & \longrightarrow & \overline{B}_X(\mathcal{A}) \\ & & \downarrow & & \\ \text{modular scalars} & \longleftarrow & \text{line operators} & \longleftarrow & \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) \end{array}$$

At the base sits a factorization category on the curve with a chosen vacuum; its endomorphism algebra is a chiral algebra; the bar construction carries the algebra to a coalgebra on compactified configurations; the centre is a three-dimensional object; its module categories carry the line operators; at the top sit the scalars — characters, Verlinde numbers, modular forms. Each floor is recovered from the next by a reconstruction theorem: Morita theory at the base, bar–cobar inversion above it, terminality of the centre above that, the modular trace at the top. One element, the Maurer–Cartan element $\Theta_{\mathcal{A}}$ of Section 5, generates every floor; one involution, Koszul duality, acts on all of them.

1.3. **The results.** Sections 2–3 construct the bar complex of a chiral algebra geometrically, as the space of logarithmic forms on Fulton–MacPherson compactifications, and prove that the construction is a selection principle: the differential squares to zero if and only if the input satisfies the Borcherds identity (Theorem 2.2). A companion rigidity result sharpens this: nilpotence consumes the full differential, level term included (Proposition 2.3); it is a cancellation between geometry and algebra, finer than any Jacobi identity.

Section 3 separates three operations: reconstruction (the bar–cobar counit), Koszul duality (Verdier duality on the Ran space applied to the bar coalgebra), and the derived centre — three functors with three targets, related exactly by the theorems that relate them. The main recognition result is a Poincaré–Birkhoff–Witt criterion (Theorem 3.2) establishing Koszulness for the universal affine, Virasoro, $\mathcal{W}$, free-field and lattice algebras, and locating its exact boundary: the Ising model and the simple admissible quotients $L_k(\mathfrak{sl}_3)$ at denominator $q \geq 3$ lie outside it, each for an identified structural reason.

Section 4 proves that the ordered bar complex is the primitive object and the symmetric one its shadow. The kernel of the symmetrisation map has dimension $d^n - \binom{n+d-1}{d-1}$ in arity n (Theorem 4.1), and what the kernel carries is precisely the quantum-group data: the R-matrix at degree two, the Drinfeld associator at degree three, the full braid monodromy beyond. Koszul duality acts on the ordered theory by inverting the noncommutativity datum, $R \mapsto R^{-1}$, realised geometrically as orientation reversal of collision cycles (Theorem 4.2); for the Yang R-matrix this gives $Y_\hbar(\mathfrak{sl}_2)^! \simeq Y_{-\hbar}(\mathfrak{sl}_2)$ (Theorem 4.3).

Section 5 passes to positive genus. The unique replacement for $z_1 - z_2$ is the prime form; the corrected propagator acquires a non-holomorphic term, and the fibrewise bar differential curves:

$$d_{\mathrm{fib}}^2 \;=\; \kappa(\mathcal{A}) \cdot \omega_g, \tag{1.3}$$

where ω_g is the first Chern class of the Hodge bundle and $\kappa(\mathcal{A})$ is a scalar computed by a two-channel residue calculation (Theorem 5.1). Flatness is restored by the Gauss–Manin connection, and since the boundary of the boundary of $\overline{\mathcal{M}}_{g,n}$ is empty, the total differential $D_{\mathcal{A}}$ satisfies $D_{\mathcal{A}}^2 = 0$ unconditionally. Consequently $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is a Maurer–Cartan element in the modular convolution Lie algebra (Theorem 5.5); this is the central object of the paper. For the algebras of central interest the completion is intrinsic — an explicit witness forces it — and the correct ambient is the coderived category of the second kind (Theorem 5.6).

Section 6 computes the obstruction tower of $\Theta_{\mathcal{A}}$ restricted to a primary line. The generating function $H(t) = \sum_r rS_r t^r$ of the tower satisfies

$$H(t)^2 \;=\; t^4\, Q(t), \qquad Q(t) = (2\kappa + 3S_3 t)^2 + 2\Delta\, t^2, \qquad \Delta = 8\kappa S_4, \tag{1.4}$$

an algebraic equation of degree two determined by three numbers (Theorem 6.1). The possible depths of the tower are 2, 3, 4 or ∞, and depth 3 on a single line is impossible (Theorem 6.2); this dichotomy is the origin of the classification of chiral algebras into four classes with a fifth critical stratum (Theorem 6.3).

Section 10 proves the complementarity law: the cohomology of the centre local system over $\overline{\mathcal{M}}_g$ splits into contributions of $\mathcal{A}$ and of its Koszul dual, and the scalar trace of the splitting is the sum $\varkappa = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!)$. For the affine and free-field families $\varkappa = 0$; for the Virasoro family $\varkappa = 13$, equivalently $c + c^! = 26$ — the ghost budget of the bosonic string arising here as arithmetic (Theorem 10.1).

Section 7 works the theory family by family — Heisenberg, affine, level one, symmetric orbifolds, $\beta\gamma$ — until every clause of the classification has a computed witness, and Section 8 crosses the landscape by Drinfeld–Sokolov reduction: gauge theory becomes gravity, class L becomes class M, and the $\mathcal{W}$-algebras acquire their quantum groups, including the chiral bialgebra of $\mathcal{W}_{1+\infty}$ with its computed antipode obstruction. Section 9 assembles the recognition theory into a moduli problem over the torsor of associators and extracts the fingerprint: five numbers that determine a landscape algebra up to the braiding.

Section 10 proves the complementarity law, the quantisation law $\hbar^2\varkappa = -1$, and the self-dual points. Section 11 computes the genus expansion. Its scalar part is a universal multiple of the Faber–Pandharipande numbers,

$$\sum_{g\geq 1} F_g^{\mathrm{sc}}\, x^{2g} = \kappa(\mathcal{A})\Big(\frac{x/2}{\sin(x/2)} - 1\Big), \tag{1.5}$$

the Wick-rotated $\widehat{A}$-genus (Theorem 11.1). The scalar formula is exact for algebras with a single generator weight and fails beyond: for $\mathcal{W}_3$ at genus two the correction is $(c+204)/16c$ (Theorem 11.2), and the corrections dominate the scalar part at large genus. The genus tower converges; the corresponding string expansion diverges factorially. The shadow is the convergent projection of a divergent theory.

Section 12 reads the ordered theory spectrally — the chain from the Maurer–Cartan element through the r-matrix and the transfer matrix to the Baxter operator, with the Bethe ansatz as its chain-level refinement. Section 13 descends to conformal blocks: the Verlinde formula from chiral homology, the Dedekind eta function as a Fredholm determinant, the Kac–Wakimoto characters from the bar resolution, and the critical level, where the centre becomes the algebra of functions on opers. Section 14 opens the genus-two window: the level enters the curvature, the double replaces the Yangian, the Igusa form enters the sewing kernel.

Section 15 identifies the closed sector. The derived chiral centre, acting on the boundary algebra through the two-coloured Swiss-cheese operad, is the terminal local open–closed pair (Theorem 15.1): every local closed sector acting on $\mathcal{A}$ factors uniquely through it. This is holography as a universal property, and it is sharp: a physical bulk theory is specified by the algebraic data together with one further datum, an open–closed comparison map, and Proposition 15.2 shows the datum is irreducible.

Section 16 exhibits the two arithmetic channels of the tower and their asymmetry: the Virasoro tower is the iterated self-extension of a single Kummer motive — infinite depth, no transcendence — while for lattice algebras the tower resolves an Epstein zeta function one Hecke eigenform per degree, so that the quartic obstruction of the Leech lattice algebra is the Ramanujan L-function. Kummer-irregular primes enter the Verlinde polynomials through Bernoulli numbers (Theorem 16.1) and provably avoid the Virasoro tower in the computed range.

Section 17 completes the local picture — the derived centre of the affine algebra is the observable algebra of quantum Chern–Simons theory on the disc, and the operadic circle $E_3 \to E_2 \to E_1 \to E_2 \to E_3$ closes — and Section 18 opens the geometric source: the Atiyah–Connes class of a Calabi–Yau category, the spine theorem carrying it to the first bar–cobar obstruction of the associated chiral algebra, and the quantum groups of $K3$. Section 19 reaches the automorphic boundary, where the strictification mechanism of the ordered theory locates the imaginary roots. Section 20 records the physics the formalism derives, each statement carrying its comparison map, and Section 21 states the one problem to which the paper reduces.Every result stated in this paper is proved in this paper, at the scope its statement declares; the monograph [Lor] develops the surrounding theory at length.

1.4. **Relation to the literature.** The objects assembled here have distinguished ancestries. Chiral algebras and the Ran space are Beilinson–Drinfeld [BD]; the categorical bar–cobar equivalence in that setting is Francis–Gaitsgory [FG]; factorization algebras as quantum field theory are Costello–Gwilliam [CG]. The compactifications are Fulton–MacPherson [FM], the relations Arnold [Ar], the modular operads and their Feynman transform Getzler–Kapranov [GK], the curved coalgebras and the second-kind derived categories Positselski [Pos]. The Hodge-integral evaluations are Mumford and Faber–Pandharipande [Mum, FP]; the prime form is Fay [Fay]; the r-matrices and quantum groups are Drinfeld [Dr], their chiral quantization Etingof–Kazhdan [EK], their geometric action Maulik–Okounkov [MO] and Costello–Witten–Yamazaki [CWY]. What is new is the single construction that holds them together: the geometric bar complex whose nilpotence is the Borcherds identity, its curvature over moduli, the Maurer–Cartan element it defines, and the computation of that element's projections across the entire standard landscape.

Conventions. We work over $\mathbb{C}$. Complexes are cohomologically graded; differentials have degree $+1$. For a graded vector space V, sV and $s^{-1}V$ denote suspension and desuspension. A chiral algebra on X is a right $\mathcal{D}_X$-module $\mathcal{A}$ with a chiral product $\mu\colon j_*j^*(\mathcal{A}\boxtimes\mathcal{A}) \to \Delta_!\mathcal{A}$ satisfying locality, unitality and the Borcherds identity, in the sense of Beilinson and Drinfeld [BD]; augmented means equipped with a splitting $\varepsilon\colon \mathcal{A}\to\omega_X$, with augmentation ideal $\bar{\mathcal{A}} = \ker\varepsilon$. Locally, for fields a, b the product is encoded by the operator product expansion $a(z)b(w) \sim \sum_{n\geq 0}(a_{(n)}b)(w)/(z-w)^{n+1}$.

2. The bar construction on configuration spaces

2.1. **Compactified configurations.** Let $\mathrm{Conf}_n(X) \subset X^n$ be the space of n distinct labelled points and let $\mathrm{FM}_n(X)$ be its Fulton–MacPherson compactification [FM]: the iterated blow-up of X^n along the partial diagonals in increasing order of dimension. It is a smooth manifold with corners whose boundary is a normal crossing divisor $D = \bigcup_S D_S$ indexed by subsets S with $|S| \geq 2$; strata are indexed by nested families of subsets, and the open part of D_S fibres over $\mathrm{FM}_{n-|S|+1}(X)$ with fibre a screen: the space of $|S|$ points on the tangent line modulo affine transformations. On a curve the pairwise diagonals are already divisors, so the screen of a pair is a point; the first genuine blow-up occurs at triple collisions.

The form (1.1) extends to $\mathrm{FM}_n(X)$ with logarithmic singularities along D; its residue along D_S equals 1 if $\{i,j\} \subseteq S$ and 0 otherwise. This residue is the only memory the compactification keeps of a collision, and it is independent of every choice. The Arnold relation (1.2) is the statement that the forms η_{ij} know about triple collisions; it is the entire codimension-two constraint on a curve.

2.2. The ordered bar complex.

Definition 2.1. Let $(\mathcal{A}, \varepsilon)$ be an augmented chiral algebra on X. The *ordered bar complex* of $\mathcal{A}$ is the cofree conilpotent tensor coalgebra

$$\overline{B}{}^{\mathrm{ord}}_X(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}) \;=\; \bigoplus_{n\geq 1} (s^{-1}\bar{\mathcal{A}})^{\otimes n} \otimes \Gamma\big(\mathrm{FM}^{\mathrm{ord}}_n(X),\, \Omega^{n-1}_{\log}\big), \tag{2.1}$$

with the deconcatenation coproduct $\Delta[a_1|\cdots|a_n] = \sum_{p=0}^{n}[a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_n]$ and the differential

$$d \;=\; d_{\mathrm{int}} + d_{\mathrm{res}} + d_{\mathrm{dR}}, \tag{2.2}$$

where d_{int} is the internal differential of $\mathcal{A}$, d_{dR} the de Rham differential of logarithmic forms, and

(2.3)
$$d_{\mathrm{res}}\big([a_1|\cdots|a_n]\otimes\alpha\big) = \sum_{i=1}^{n-1}\sum_{r\geq 0}(-1)^{\epsilon_i}\,[a_1|\cdots|a_{i(r)}a_{i+1}|\cdots|a_n] \otimes \mathrm{Res}_{D_{i,i+1}}\alpha,$$

with $\epsilon_i = \sum_{q<i}(|a_q| - 1) + |a_i|$. The sum in (2.3) runs over *all* polar modes of the operator product: the logarithmic kernel absorbs one order of pole, so a pole of order $r+1$ in the OPE contributes the mode $a_{(r)}b$.

The differential extracts consecutive collisions only; this is what the word *ordered* means, and the boundary faces that appear are exactly the $n-1$ consecutive-pair divisors, the facets of the Stasheff associahedron. The homotopy structure is integral-geometric: realising the associahedron K_n as the real ordered compactification inside the complex one, the transferred A_∞ operations are the integrals $m_n = \int_{K_n} Y(a_1, z_1)\cdots Y(a_n, z_n)\,\omega_n$, the Stasheff identities are the cancellation of the codimension-one facets, and the pentagon first appears at codimension three — coherence is Stokes' theorem, applied to a polytope. The symmetric bar complex $\overline{B}^{\Sigma}_X(\mathcal{A})$ is the Σ_n-coinvariant quotient, a factorization coalgebra on the Ran space of X; the relation between the two is the subject of Section 4.

2.3. **The selection principle.**

Theorem 2.2 (Selection principle)**.** *Let $\mathcal{A}$ be an augmented $\mathcal{D}_X$-module equipped with binary operations $a_{(r)}b$ of all orders. The operator d of* (2.2) *satisfies $d^2 = 0$ if and only if the operations satisfy the Borcherds identity. The two constraints decouple by stratum: on codimension-two boundary strata with disjoint supports, residues anticommute for orientation reasons; on triple-collision strata the cyclic sum of iterated residues vanishes if and only if the Borcherds identity holds, the geometric coefficient being supplied by the Arnold relation* (1.2)*.*

Proof. Stokes' theorem on the manifold with corners $\mathrm{FM}_n(X)$ reduces d^2 to a sum over codimension-two boundary strata, and the strata sort into three types. *Disjoint pairs*: a stratum $D_{ij} \cap D_{kl}$ with $\{i,j\} \cap \{k,l\} = \varnothing$ is reached through its two codimension-one parents in either order; the two iterated residues carry opposite orientation signs, from the anticommutation $e_{ij} \wedge e_{kl} = -e_{kl} \wedge e_{ij}$ of the conormal directions, and cancel pairwise. *Repeated pairs*: a single divisor contributes one Poincaré residue only, the first extraction consuming the unique logarithmic normal factor, so no second-order term arises. *Triple collisions*: the stratum over D_{ijk} is approached through its three parents D_{ij}, D_{jk}, D_{ik}; the three iterated residues assemble, by the Arnold relation (1.2) applied to the form factors, into the cyclic sum of compositions $(a_{(m)}b)_{(n)}c$ over all admissible mode orders, and the vanishing of this sum for all modes and all c is precisely the Borcherds identity. The geometry supplies the coefficient, the algebra the cancellation; each is used exactly once. Conversely, a triple of modes violating the identity produces a chain with $d^2 \neq 0$ supported on the corresponding stratum, so the implication is an equivalence; the orientation bookkeeping is completed in [Lor]. □

Theorem 2.2 says the bar construction *recognises* chiral algebras: the Borcherds identity is the exact condition the geometry accepts. Configuration-space geometry accepts exactly the algebras satisfying the Borcherds identity. The following negative result shows how little of the differential is dispensable.

Proposition 2.3. *Let $\mathcal{A} = V_k(\mathfrak{sl}_2)$ and let d_{br} denote the part of d_{res} retaining only the simple-pole modes $a_{(0)}b$. Then $d_{\mathrm{br}}^2 \neq 0$, for every one of the 2^{11} admissible assignments of signs to the terms of d_{br}. Nilpotence of the full differential requires the double-pole (level) term.*

Nilpotence consumes the level term: the central extension is structure, spent in the proof of $d^2 = 0$.

2.4. **First computations.** Write B^n for the bar complex in bar degree n and H^n for its cohomology, on $X = \mathbb{P}^1$ at a fixed point of the augmentation.

Example 2.4 (Heisenberg)**.** For the rank-one Heisenberg algebra $\mathcal{H}_k$, generated by α with $\alpha_{(1)}\alpha = k$, the residue differential vanishes on words in α except through the double pole, which produces a scalar: the bar complex is quadratic, the cohomology is $(\mathbb{C},\, 0,\, \mathbb{C}\cdot c_k,\, 0, \dots)$, and the class in degree two is the central extension of the loop algebra. The dual object is a curved symmetric coalgebra with curvature $-k\omega$; it is *not* the Heisenberg algebra at level $-k$.

Example 2.5 (Free fermion)**.** For the free fermion the cyclic group $\mathbb{Z}/3$ acts on the two-dimensional space $\Omega^2_{\log}(\mathrm{FM}_3)$ with eigenvalues the primitive cube roots of unity; total antisymmetry of $\psi \otimes \psi \otimes \psi$ forces invariance; the intersection is zero and the bar cohomology collapses to $\mathbb{C}$.

Example 2.6 (Affine Kac–Moody)**.** For $\widehat{\mathfrak{g}}_k$ the bar chain space in degree n has dimension $(\dim \mathfrak{g})^n\, (n-1)!$ — the factor $(n-1)!$ is the rank of $\Omega^{n-1}_{\log}$, the Arnold–Orlik–Solomon count — and the level enters only the differential, not the chain space. For $\widehat{\mathfrak{sl}}_2$ the cohomology has dimension $2n+1$ in degree n, with generating function $x(3-x)/(1-x)^2$. The cohomology is Chevalley–Eilenberg cohomology of the loop algebra corrected by the Orlik–Solomon factor, and the factor is exact: dimension $2n+1$ in degree n.

Example 2.7 (Virasoro)**.** Every presentation of the Virasoro algebra carries a quartic pole: the operator product of T with itself has a fourth-order pole and T appears on both sides. Its bar cohomology has dimensions given by first differences of the Motzkin numbers, with generating function $4x/\big(1-x+\sqrt{1-2x-3x^2}\,\big)^2$. In degree two, the vacuum term $(c/2)[T]\otimes(\eta_{12}+\eta_{13}+\eta_{23})$ survives: the Arnold relation is quadratic, not linear, and the surviving class is the germ of the curvature of Section 5.

2.5. **Global sections and boundaries.** Two global statements complete the local picture. First, the homology of the geometric bar complex computes factorization homology: for holonomic $\mathcal{A}$, Weiss excision identifies $H_*\big(\overline{B}^{\mathrm{geom}}(\mathcal{A})\big) \cong \int_X \mathcal{A}$, so the bar construction is a chain-level model for the global invariant, with the coradical filtration realised geometrically by the depth of collision trees. Second, on a curve with marked points and tangent directions the correct compactification is the bordered one: real oriented blow-up replaces punctures by boundary circles, and the codimension-one boundary of the fixed-curve moduli has exactly three types of face — interior collision, carrying the screens above; consecutive boundary collision, carrying Stasheff associahedra and extracting A_∞ operations on boundary modules; and mixed bubbling, carrying the Swiss-cheese operad and extracting bulk-to-boundary maps. The fourth type of face, nodal degeneration of the curve itself, exists only in the relative situation over $\overline{\mathcal{M}}_{g,n}$ and is the subject of Section 5. The three fixed-curve face types are the complete geometric source of the open–closed structure of Section 15, complete as it stands.

2.6. **Modules.** Modules enter through pointed configurations: the module bar complex of an $\mathcal{A}$-module M lives on the compactification with one marked point, is a comodule over $\overline{B}_X(\mathcal{A})$ by deconcatenation at the marked point, and its degree-zero cohomology with several insertions is the space of derived coinvariants whose comparison with conformal blocks occupies Section 13. Every duality of this paper has a module form, obtained by marking a point in its proof.

3. Reconstruction, duality, and the centre

3.1. Three functors.

Definition 3.1 (Five objects)**.** Five objects are associated to an augmented chiral algebra $\mathcal{A}$, in five different categories:

(1) the algebra $\mathcal{A}$ itself;
(2) the bar coalgebra $\overline{B}_X(\mathcal{A})$;
(3) for a quadratic presentation $\mathcal{A} = T_X(V)/(R)$, the quadratic coalgebra $\mathcal{A}^i = C_X(s^{-1}V, s^{-2}R)$ with its canonical comparison $q_{\mathcal{A}}\colon \mathcal{A}^i \to \overline{B}_X(\mathcal{A})$;
(4) the Verdier algebra $K_X(\mathcal{A}) = \mathbb{D}_{\mathrm{Ran}}\,\overline{B}_X(\mathcal{A})$, the Verdier dual of the bar coalgebra as a holonomic object on the Ran space, together with a chosen pair map $\nu_{\mathcal{A}}\colon K_X(\mathcal{A}) \to \mathcal{A}^! = (\mathcal{A}^i)^\vee$;
(5) the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = R\mathrm{Hom}_{\mathcal{A}\otimes\mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A})$.

The counit $\Omega_X\overline{B}_X(\mathcal{A}) \to \mathcal{A}$ of the bar–cobar adjunction is *reconstruction*: a covariant operation with value $\mathcal{A}$. Koszul duality is the contravariant operation (4). The centre is the third operation (5). An equivalence between the covariant and the contravariant outputs is precisely a self-duality datum on $\mathcal{A}$.

The adjunction itself is classical in the enhanced setting: for the reduced associative operad in the pro-nilpotent Ran ambient of Francis and Gaitsgory the bar and cobar functors are mutually inverse equivalences between augmented algebras and conilpotent coalgebras [FG]. What requires proof in the present geometric model is, first, that the chain-level realisation on Fulton–MacPherson spaces computes this equivalence — this holds on the completed finite-type locus, by a Mittag–Leffler argument recalled in Section 5 — and, second, when the quadratic comparison $q_{\mathcal{A}}$ is a quasi-isomorphism. The latter is Koszulness.

3.2. **The recognition theorem.**

Theorem 3.2 (PBW criterion)**.** *Let $\mathcal{A}$ carry an exhaustive filtration $F_\bullet$ with the properties: F is flat; the associated graded $\mathrm{gr}_F\mathcal{A}$ is a classically Koszul commutative chiral algebra; each finite conformal-weight window is finite-dimensional and the resulting towers are Mittag–Leffler. Then $q_{\mathcal{A}}$ is a quasi-isomorphism: $\mathcal{A}$ is chirally Koszul. In particular the universal affine algebra $V_k(\mathfrak{g})$ for every level k, the Virasoro algebra Vir_c for every c, the universal principal $\mathcal{W}$-algebras, the free-field algebras and the lattice algebras are chirally Koszul.*

Proof. Filter the twisted tensor complex by PBW degree. Flatness identifies the associated graded differential with the classical Koszul complex of $\mathrm{gr}_F\mathcal{A}$, which is acyclic by Priddy's theorem; the bracket and level terms are of lower order. The spectral sequence collapses at E_1; Mittag–Leffler gives strong convergence: the conformal-weight windows are finite-dimensional, their tower has surjective transitions, and the limit closes by Mittag–Leffler. $\square$

Koszulness is a property of the universal algebra, not of its quotients:

Theorem 3.3. *The Ising chiral algebra $M(3,4)$ lies outside the Koszul locus: the null-vector quotient removes a bar-degree-one chain at weight $(p-1)(q-1)$ while its image in degree two survives, producing $H^2(\overline{B}) \neq 0$ off the diagonal. The simple admissible quotients $L_k(\mathfrak{sl}_3)$ at denominator $q \geq 3$ lie outside it as well: the Cartan subalgebra contributes $\mathrm{rk}(\mathfrak{g})$ permanent classes to $H^2(\overline{B})$. Koszulness is logically independent of C_2-cofiniteness: $V_k(\mathfrak{g})$ is Koszul and generically not C_2-cofinite, while C_2-cofinite simple quotients may fail Koszulness.*

On the recognition locus the property has many faces. Six conditions are equivalent through the single hub of PBW collapse: the existence of a Koszul twisting datum; $q_{\mathcal{A}}$ a quasi-isomorphism; the cobar comparison $\Omega_X(\mathcal{A}^i) \to \mathcal{A}$ a quasi-isomorphism; acyclicity of the two twisted tensor complexes; collapse of the bar spectral sequence on the diagonal; and diagonal concentration of $\mathrm{Ext}_{\mathcal{A}}(\mathbf{1},\mathbf{1})$. A seventh condition, formality over the Swiss-cheese operad, is finer: it is equivalent to the conjunction of the hub with membership in the Gaussian class of Section 6, and the affine and Virasoro algebras, Koszul and non-Gaussian, exhibit the gap between the six and the seventh.

3.3. **Curvature of the dual.** The dual of an algebra whose OPE has double poles is curved. For $\mathcal{H}_k$ the dual is $(\mathrm{Sym}^{\mathrm{ch}}(V^*[1]), m_0 = -k\omega)$; the curvature m_0 is the level. Curved coalgebras are objects of Positselski's coderived category of the second kind [Pos]: the derived category of comodules taken with respect to coacyclic rather than acyclic objects, in which a curved bar–cobar comparison with coacyclic cone is an equivalence and the comodule–contramodule correspondence $D^{\mathrm{co}}(C\text{-comod}) \simeq D^{\mathrm{ctr}}(C\text{-contra})$ holds for any coalgebra with finite-dimensional graded pieces. Four regimes organise the inversion theory, each with its own convergence mechanism:

regime	*examples*	*curvature*	*mechanism*
quadratic	Heisenberg, lattice	0	none needed
curved central	affine, Virasoro	$\kappa\,\omega_g$	coderived category
filtered complete	$\mathcal{W}_N$	central	adic completion
infinite type	$\mathcal{W}_\infty$	central	weight-completed towers

The distinction becomes structural at positive genus and is taken up in Section 5.

3.4. **Completed inversion.** The chain-level realisation of the bar–cobar equivalence holds in four regimes, each with its own convergence mechanism, and the passage between them is a single lemma applied four times.

Theorem 3.4 (Completed inversion)**.** *Let $\mathcal{A}$ carry a complete descending filtration whose finite quotients $\mathcal{A}_{\leq N}$ are finite-type, with surjective transition maps. Then the completed bar and cobar functors exist with continuous differentials, and the completed counit $\widehat{\Omega}\,\widehat{\overline{B}}(\mathcal{A}) \to \mathcal{A}$ is a quasi-isomorphism in the square-zero total model: a bar word of total weight w involves only modes bounded by w, so each finite window stabilises; the Milnor exact sequence $0 \to \varprojlim^1 H^{m-1} \to H^m(\widehat{\ }) \to \varprojlim H^m \to 0$ then closes the limit, the $\varprojlim^1$ term dying on the surjective tower. In the genuinely curved regime the same finite-stage argument yields a coacyclic-cone comparison in the coderived category rather than a quasi-isomorphism. The four regimes — quadratic (Heisenberg, lattice: no completion needed), curved-central (affine, Virasoro: scalar curvature), filtered-complete ($\mathcal{W}_N$: adic completion), and infinite-type ($\mathcal{W}_\infty$: weight-completed towers) — are distinct, and every completed statement in this paper declares its regime.*

3.5. **Modules and the character identity.** On the quadratic genus-zero lane, Koszul duality extends to modules: the categories of complete modules over $\mathcal{A}$ and conilpotent comodules over $\overline{B}(\mathcal{A})$ are equivalent, with Ext exchanged; the numerical shadow of the equivalence is the Koszul character identity

$$h_{\mathcal{A}}(t)\, h_{\mathcal{A}^i}(-t) \;=\; 1. \tag{3.1}$$

For the rank-d Heisenberg algebra with its Fock modules the exchange is completely explicit: distinct Fock modules have no extensions, and the bigraded Ext series $\prod_n (1+tq^n)^d$ returns $\eta(\tau)^d$ at $t=-1$ — the character identity meeting the Fredholm determinant of Section 13.

4. The ordered theory and quantum groups

4.1. What averaging forgets. The symmetric bar complex is the quotient of the ordered one by the action of the symmetric groups. The quotient map is the Reynolds averaging operator, and it is lossy in a way that can be computed exactly.

Theorem 4.1 (Reynolds kernel)**.** *Let V be a d-dimensional space of fields. The kernel of the averaging map $\mathrm{av}_n\colon V^{\otimes n} \to \mathrm{Sym}^n V$ has dimension*

$$\dim\ker(\mathrm{av}_n) \;=\; d^n - \binom{n+d-1}{d-1}, \tag{4.1}$$

with generating function $\frac{1}{1-dt} - \frac{1}{(1-t)^d}$: the ordered theory grows exponentially, its symmetric shadow polynomially.

Proof. Schur–Weyl decomposition of $V^{\otimes n}$ against the stars-and-bars count of $\mathrm{Sym}^n V$; the Reynolds operator projects each isotypic block onto its trivial part, and every non-trivial Σ_n-isotypic lies in the kernel. □

The content of the kernel is graded by degree. At degree two the ordered theory retains the binary collision residue

$$r_{\mathcal{A}}(z) \;=\; \mathrm{Res}^{\mathrm{coll}}_{0,2}\big(\Theta^{E_1}_{\mathcal{A}}\big), \tag{4.2}$$

the classical r-matrix; averaging retains only its trace, the scalar κ of Section 5, with a shift: for the affine algebra the simple-pole self-contraction through the adjoint Casimir adds $\dim\mathfrak{g}/2$, so that $\kappa = \mathrm{av}(r) + \dim\mathfrak{g}/2 = \dim\mathfrak{g}\,(k + h^\vee)/2h^\vee$. At degree three the kernel contains the commutator $[\Omega_{12}, \Omega_{23}]$ and with it the Drinfeld associator: any splitting of the averaging map is a choice of associator, and the set of splittings is a torsor over the Grothendieck–Teichmüller group $\mathrm{GRT}_1(\mathbb{Q})$. At genus one a further obstruction appears: the modular completion of the propagator replaces E_2 by $E_2 - 3/(\pi\,\mathrm{Im}\,\tau)$, and the non-holomorphic correction lives strictly beyond the holomorphic theory; the ordered and symmetric theories are bound together over the modular parameter, the splitting obstruction being exactly this anomaly. Beyond degree three the kernel carries the full pure-braid monodromy of the Knizhnik–Zamolodchikov connection: individual conformal blocks as vectors, of which the symmetric theory keeps only the count.

Quantum groups, in one sentence, are the E_1 memory of chiral algebras.

4.2. The inversion principle. Koszul duality acts on the ordered theory through Verdier duality on ordered configuration spaces, and Verdier duality reverses the orientation of collision cycles. Orientation reversal inverts the noncommutativity datum.

Theorem 4.2 (Inversion principle)**.** *On the ordered theory, Koszul duality inverts the braiding datum: for a lattice algebra with 2-cocycle ε the dual carries ε^{-1}; for a braided input with R-matrix $R(z)$ the dual carries $R(z)^{-1}$; on quantum parameters, $q \mapsto q^{-1}$. The operadic form of the statement is* $(\mathrm{Ass}^{\mathrm{ch}})^! \cong \mathrm{Ass}^{\mathrm{ch}} \otimes \mathrm{sgn}$.

Proof. Verdier duality exchanges the j_*-extension with logarithmic forms and the $j_!$-extension with boundary currents, and on ordered configuration spaces it reverses the co-orientation of each collision cycle. The pairing of a cycle with the noncommutativity datum — cocycle value, R-matrix, braiding phase — is therefore inverted, and the sign representation records the reversal at the operadic level. □

This has a consequence worth isolating. The level inversion $k \mapsto -k - 2h^\vee$ of the symmetric theory (the unique affine involution fixing the critical level and compatible with κ) and the braiding inversion $q \mapsto q^{-1}$ of the ordered theory are the same operation seen at two depths, since $q = e^{\pi i/(k+h^\vee)}$.

4.3. **The Yangian.** The Yangian enters not as an example but as the value of the ordered duality functor. In the RTT presentation, with Yang matrix $R_\hbar(u) = 1 - \hbar P/u$, the defining relation $R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v)$ equips the completed Yangian with a quadratic E_1-factorization structure whose braiding is genuinely non-symmetric: its bar complex exists only on ordered configurations.

Theorem 4.3. *For the Yang kernel in type A, the orthogonal complement of the RTT relation space under the trace pairing is the relation space of the inverse kernel, and $R_\hbar(u)^{-1} = (1 - \hbar^2/u^2)^{-1}R_{-\hbar}(u)$. Each finite RTT window therefore satisfies $A_N(R_\hbar)^! \cong A_N(R_{-\hbar})$, and since a bar word of total weight w involves only modes $\leq w$, the windows stabilise and the completed statement follows:*

$$Y_\hbar(\mathfrak{sl}_2)^! \;\cong\; Y_{-\hbar}(\mathfrak{sl}_2), \qquad \hbar = \frac{1}{k+h^\vee}. \tag{4.3}$$

Proof. The RTT relation presents the window as the image of the operator $X \mapsto R_{12}X - \sigma(X)R_{12}$; a direct computation in the fundamental square identifies the orthogonal complement of this image under the trace pairing with the corresponding image for R^{-1}, and unitarity $R_{12}(u)R_{21}(-u) = (\hbar^2 - u^2)\,\mathrm{id}$ converts R^{-1} into $R_{-\hbar}$ up to the central scalar, which preserves the relation span. Quadratic duality on each window is therefore the sign flip; the windows stabilise as stated, and the Milnor sequence completes the limit. □

Three remarks fix the scope. First, the identification of the ordered Koszul dual of $\widehat{\mathfrak{g}}_k$ itself with the Yangian requires, beyond Theorem 4.3, a strictification of the quasi-associative structure; the obstruction lives in root spaces and vanishes when all root multiplicities are one — the mechanism fails precisely for Kac–Moody algebras with imaginary roots, which is where the Borcherds algebras of Section 16 live. Second, the ordered double dual loses the level: $(\mathcal{A}^!)^! \simeq U(\mathfrak{g}[t])$, not $\widehat{\mathfrak{g}}_k$. The central extension is held by the closed (holomorphic) colour through the double pole, the Yangian datum by the open (topological) colour through the simple pole; only the two-coloured double duality is involutive. The two colours of the Swiss-cheese operad carry disjoint information, each indispensable. Third, at genus one the B-cycle monodromy of the elliptic propagator exponentiates $\hbar$ to $q = e^{\pi i/(k+h^\vee)}$; this is a monodromy statement about local systems; the algebra isomorphism with $U_q(\mathfrak{g})$ is a separate statement, and monodromy is what the geometry supplies.

4.4. **Monodromy, Langlands duality, and the evaluation core.** The degree-three content of the ordered theory is monodromy. The shadow connection of the affine algebra is the Knizhnik–Zamolodchikov connection; its monodromy generates the braid-group representations of the quantum group at $q_{\mathrm{KL}} = e^{\pi i/(k+h^\vee)}$, and the level reflection acts as $q \mapsto q^{-1}$, which is Theorem 4.2 again. The identification of the quantum parameter is overdetermined: for $\widehat{\mathfrak{sl}}_2$ the braiding eigenvalues on the two channels of the fundamental square are $-q^{-3/2}$ and $q^{1/2}$, and four independent computations — the KZ monodromy, the evaluation-module braiding at $u = \cot \frac{\pi}{k+2}$, the fusion-category twist, and the B-cycle monodromy at genus one — return the same q; the number $1/(k + h^\vee)$ is the unique coupling compatible with the Sugawara construction, and every road leads to it. Three statements calibrate what is known beyond type A.

First, the categorical Clebsch–Gordan input holds in complete generality: for every simple $\mathfrak{g}$, the tensor product of a fundamental evaluation module with a negative prefundamental in the shifted category [HJ] $\mathcal{O}$ splits according to the q-character, $V_{\omega_i}(a) \otimes L_i^-(b) \cong \bigoplus_\mu L_i^-(\text{shift} = \mu)$, at generic parameters: the summand ℓ-weights are multiplicity-free and pairwise unlinked, so the filtration by singular vectors splits. On the category generated by evaluation modules, ordered Koszul duality is therefore established for all simple types; its extension from the evaluation core to the full category is a completion problem, open beyond type A, and in type A reduced to a single compact-generation statement on the Baxter locus $b = a - \frac{\lambda+1}{2}$.

Second, the Langlands form of the duality is exactly as strong as its inputs: granting the lacing isomorphism of Frenkel–Hernandez [FH] and a Chevalley lift of $\hbar \mapsto -\hbar$, the inverse-kernel duality of Theorem 4.3 transports to $Y_\hbar(\mathfrak{g})^! \simeq Y_{-\hbar\, r_{\mathfrak{g}}}(\mathfrak{g}^L)$, with $r_{\mathfrak{g}}$ the lacing number: self-dual for the simply-laced types, and exchanging B_r with C_r. For the exceptional types the duality is a finite construction in seven items — kernel, projectors, Poincaré–Birkhoff–Witt basis, relation-space isomorphism, scalar gauge, tower compatibility, chiralisation — whose completion yields $\widehat{Y}_\hbar(\mathfrak{g})^! \cong \widehat{Y}_{-\hbar}(\mathfrak{g})$. The complementarity equation $\kappa(V_k(\mathfrak{g})) + \kappa(V_{k^\vee}(\mathfrak{g}^\vee)) = 0$ has the unique solution $k^\vee = -k - 2h^\vee$: the level of the dual theory is fixed by the vanishing of the total curvature.

Third, the super form: for $Y_\hbar(\mathfrak{sl}(m|n))$ with $m \neq n$ the supertrace complementarity $\kappa^{\mathrm{str}} + \kappa^{\mathrm{str},!} = 0$ holds under $k \mapsto -k - 2(m - n)$; the Berezinian-normalised sum is $\max(m, n)$: a conjecture, supported by the closed forms at low rank.

4.5. **The Etingof–Kazhdan triangle.** On the ordered Koszul locus, at a fixed associator, three structures on the bar coalgebra determine one another: a vertex R-matrix $S(z)$ satisfying the Etingof–Kazhdan axioms; the curved A_∞-structure of the ordered bar complex; and a chiral coproduct, coassociative up to the associator, on the double. The equivalence triangle is verified in full for the $\mathfrak{sl}_2$ Yangian and the affine algebras, and the quantization functor of Etingof–Kazhdan [EK] is, in this language, a choice of splitting of the averaging map — which is why its ambiguity is the same Grothendieck–Teichmüller torsor. The triangle is the precise sense in which a chiral algebra with a braiding *is* a quantum group: three definitions, one object, one torsor of identifications.

4.6. **Two normalisations.** The collision residue of the affine algebra is $r = k\Omega_{\mathrm{tr}}/z$ in the trace form; the Knizhnik–Zamolodchikov connection carries $\Omega/\big((k+h^\vee)z\big)$ with $\Omega = 2h^\vee\,\Omega_{\mathrm{tr}}$. Their ratio is $k(k+h^\vee)/2h^\vee$: no rescaling independent of the level identifies them, as evaluation at $k=0$ shows. The first vanishes at $k=0$ and is finite at the critical level; the second is the reverse. Every formula in this paper declares which of the two it uses.

4.7. **The dictionary.** The division of labour between the two theories is exact:

the symmetric theory keeps	*the ordered theory keeps*
the modular characteristic κ	the r-matrix $r(z)$
the shadow tower S_r	the associator Φ
the genus expansion F_g	the braid monodromy
characters and Verlinde numbers	conformal blocks as vectors
the conductor $\varkappa$	the quantum group

Every left-hand entry is the average of its right-hand neighbour, and each row of the left column is computed in this paper from the row on the right.

5. Higher genus: curvature and the Maurer–Cartan element

5.1. **The prime form and the curved differential.** On a curve of genus $g \geq 1$ the prime form replaces the difference $z_1 - z_2$: it is its $E(z_1, z_2)$, a section of $K^{-1/2} \boxtimes K^{-1/2}$ vanishing simply on the diagonal, and the propagator becomes

$$\eta_{12}^{(g)} = d\log E(z_1, z_2) + \pi \sum_{j,k=1}^{g} (\operatorname{Im}\Omega)^{-1}_{jk} \operatorname{Im}\Big(\int_{z_2}^{z_1} \omega_j\Big)\,\omega_k(z_1), \tag{5.1}$$

where Ω is the period matrix. The correction term is forced: the prime form is quasi-periodic around B-cycles, and (5.1) is the unique completion of $d\log E$ that is single-valued. Single-valuedness exchanges holomorphy for curvature.

Theorem 5.1 (Curvature)**.** *Let $\mathcal{A}$ be a chiral algebra of the standard landscape and let d_{fib} be the bar differential built from (5.1) on a family of genus-g*

curves. Then d_{fib} is a curved differential: $d_{\mathrm{fib}}^2 = [m_0^{(g)}, -]$ for a curvature element $m_0^{(g)}$ whose scalar diagonal is

$$\operatorname{tr}_{\mathrm{diag}}\big(m_0^{(g)}\big) = \kappa(\mathcal{A}) \cdot \omega_g, \qquad \omega_g = c_1(\mathbb{E}_g) \tag{5.2}$$

the first Chern class of the Hodge bundle. The scalar $\kappa(\mathcal{A})$ is computed at genus one by two channels: the B-cycle monodromy defect $-2\pi i$ of the propagator acting through the double pole, and a one-loop self-contraction through the simple pole with symmetry factor $\frac{1}{2}$ and adjoint Casimir $2h^\vee$. The resulting values are

$$\begin{aligned} &\kappa(\mathcal{H}_k) = k, && \kappa\big(V_k(\mathfrak{g})\big) = \frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}, && \kappa(\mathrm{Vir}_c) = \tfrac{c}{2}, \\ &\kappa(\mathcal{W}_N) = c\,(H_N - 1), && \kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1, && \kappa(V_\Lambda) = \operatorname{rk}\Lambda, \end{aligned} \tag{5.3}$$

with $H_N = \sum_{j\le N} 1/j$, and $\kappa(bc_\lambda) = -\kappa(\beta\gamma_\lambda)$; κ is additive under tensor product. At the critical level $k = -h^\vee$ the two channels cancel: $\kappa = 0$.

Proof at genus one. Two computations. The corrected propagator on the torus is $\big[\theta_1'/\theta_1(z) + 2\pi i \operatorname{Im} z/\operatorname{Im}\tau\big]dz$; it is 1-periodic and acquires the B-cycle defect $-2\pi i$. Squaring the differential, the double-pole channel transports the level around the B-cycle and produces $k\dim\mathfrak{g}\cdot\omega_1$ from the defect; the simple-pole channel closes a self-contraction whose Casimir is the Killing form, $\sum f^{ac}_{\ d} f^{bc}_{\ d} = 2h^\vee\kappa^{ab}$, with symmetry factor $\frac{1}{2}$, producing $h^\vee\dim\mathfrak{g}\cdot\omega_1$. Normalising by $2h^\vee$ gives $\kappa = (k+h^\vee)\dim\mathfrak{g}/2h^\vee$. The identity $\partial_\tau\log\eta = \frac{\pi i}{12}E_2$ identifies the scalar class with the quasi-modular anomaly of E_2, which is the analytic form of the statement that the curvature is intrinsically quasi-modular. The remaining rows of (5.3) are the same two-channel computation with the appropriate Casimir data; additivity is manifest channel by channel. □

The two channels of the proof define a canonical decomposition $\kappa = \kappa_{\mathrm{dp}} + \frac{1}{2}\dim\mathfrak{g}$: the double-pole part is the averaged r-matrix, the simple-pole part the Sugawara shift, and the level reflection $k \mapsto -k - 2h^\vee$ acts on the pair as the shear fixing the shift and negating the total — κ is the (-1)-eigenvector of the reflection, which is how the dual level in Theorem 10.1 is found. Two refinements of Theorem 5.1 deserve record. At genus one the double-pole channel alone contributes, since holomorphic differentials are constant; from genus two onward $\partial_w\omega_j \ne 0$ and the level enters the curvature directly. And the simply-laced level-one algebras collapse: their root currents are composites of $\operatorname{rk}\mathfrak{g}$ Heisenberg fields, the simple-pole channel dies, and $\kappa(\widehat{\mathfrak{g}}_1) = \operatorname{rk}\mathfrak{g}$ — the unique crossing between classes in the classification of Section 6.

Theorem 5.2 (Fay's identity as the higher-genus Borcherds identity)**.** *On a smooth curve of genus g, the vanishing of the square of the period-corrected differential D_g reduces, on triple-collision strata, to the trisecant identity of Fay for the prime form and the Szegő kernel: writing $\eta^{(g)} = h + a$ for the* *meromorphic and real-analytic parts of* (5.1)*, the cyclic sum* $\sum_{\mathrm{cyc}} h_{12}h_{23}$ *is killed by the differentiated trisecant identity, the pure* aa *terms vanish as* $(2,0)$*-forms on a curve, and the cross terms assemble the Bergman kernel, whose diagonal residue is the curvature of Theorem 5.1. The Borcherds identity governs the algebra at every genus; Fay's identity governs the geometry, and the two meet only through the scalar* κ*.*

Proof. Squaring D_g and sorting by type: the meromorphic square is the cyclic sum $\sum_{\mathrm{cyc}} h_{12}h_{23}$, which the differentiated trisecant identity of Fay [Fay] expresses through even Weierstrass data and antisymmetric form factors, hence kills; the $a\cdot a$ terms are $(2,0)$-forms on a curve and vanish; and $\bar\partial a$ is the Bergman kernel, so the cross terms reduce to the diagonal residue evaluated in Theorem 5.1. □

Theorem 5.3 (Locality)**.** *Collision residues see only the logarithmic kernel: the residue of the second term of* (5.1) *along any diagonal vanishes, so the bar differential restricted to any fixed smooth curve equals its genus-zero form. Every algebraic invariant of the bar complex — Koszulness, the obstruction tower, the classification — is genus-zero data. The genuinely modular content of a chiral algebra is concentrated in the single object constructed next.*

5.2. **The Maurer–Cartan element.** Flatness is restored by geometry. The Gauss–Manin connection of the family cancels the curvature, $D_g = d_{\mathrm{fib}} + \nabla^{\mathrm{GM}}$ satisfies $D_g^2 = 0$, and the bar complexes over all genera assemble into an algebra over the Feynman transform of the modular operad $\{\overline{\mathcal{M}}_{g,n}\}$ of Getzler and Kapranov [GK].

Definition 5.4. The *modular convolution Lie algebra* of $\mathcal{A}$ is

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \prod_{2g-2+n>0} \mathrm{Hom}_{\Sigma_n}\big(C_*(\overline{\mathcal{M}}_{g,n}),\, \mathrm{End}_{\mathcal{A}}(n)\big), \tag{5.4}$$

with differential induced by the boundary of $\overline{\mathcal{M}}_{g,n}$ and bracket by one-edge sewing; the weight filtration by $2g-2+n$ makes it pronilpotent.

Theorem 5.5 (The master object)**.** *Let* $D_{\mathcal{A}}$ *be the total genus-completed bar differential and* d_0 *its tree-level part. Then* $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ *satisfies*

$$D\,\Theta_{\mathcal{A}} + \tfrac{1}{2}\,[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0 \tag{5.5}$$

identically: the equation is the expansion of $D^2_{\mathcal{A}} = 0$*, which holds because the boundary of the boundary of* $\overline{\mathcal{M}}_{g,n}$ *is empty. No equation holds term by term, and convergence of* $\Theta_{\mathcal{A}}$ *in the weight completion is automatic.*

Theorem 5.5 is the central statement of the paper, and its proof is its point: the Maurer–Cartan equation of two-dimensional chiral physics is the transcription of $\partial^2 = 0$ on Deligne–Mumford space. Everything constructed below — the obstruction tower, the complementarity law, the genus expansion, the r-matrix, the Knizhnik–Zamolodchikov connection — is a projection of $\Theta_{\mathcal{A}}$ along a filtration. The Maurer–Cartan equation at (g,n), expanded, reads: internal differential, plus separating clutching (the bracket), plus non-separating self-sewing (a Batalin–Vilkovisky operator $\hbar\Delta$), equals zero; it determines $\Theta^{(g,n)}$ from $(0,3)$ and $(0,4)$ by induction, uniquely up to gauge on the Koszul locus. The induction replaces the sum over topologies of three-dimensional gravity; there is no divergent sum to regularise.

For the affine algebras the element is strict in closed form: on the one-channel lane $\Theta^{\mathrm{str}} = \kappa \cdot \mu \otimes \Lambda$, with μ the invariant functional and $\Lambda = \sum_{g\geq 1}(-1)^g\lambda_g$ the total Hodge class — closed because the Killing three-cocycle is closed, with vanishing bracket by the Jacobi identity, exact at every genus. The simplest gravitational background of the theory is the level times the total Chern class of the Hodge bundle, in closed form.

5.3. **The two levels of the master equation.** The proof of Theorem 5.5 lives at two levels, and separating them is part of the statement. At the convolution level, $D^2 = 0$ is pure topology: the differential is composition against the singular boundary of $\overline{\mathcal{M}}_{g,n}$, and its square vanishes because $\partial^2 = 0$; no geometry of the fibres enters. At the ambient level — where the differential carries, besides clutching, the collision faces, the logarithmic degenerations, and the planted-forest corrections of the relative compactification — the square-zero identity is the signed residue–pushforward calculus of the logarithmic Fulton–MacPherson geometry, and it holds granted that calculus; the fixed-smooth-curve core stands on its own. Every scalar consequence drawn in this paper factors through the convolution level.

Expanded, the master equation is the quantum master equation of Batalin and Vilkovisky. The separating boundary is the bracket; the non-separating boundary is a second-order operator Δ raising genus and lowering arity by two; and (5.5) reads $\hbar\Delta S + \frac{1}{2}\{S,S\} = 0$ for $S = \Theta_{\mathcal{A}}$. The clutching maps factorise the element, $\xi_h^*\Theta^{(g)} = \sum_{g_1+g_2=g}\Theta^{(g_1)}\star\Theta^{(g_2)}$ and $\xi_{\mathrm{irr}}^*\Theta^{(g+1)} = \Delta\,\Theta^{(g)}$, and Verdier duality intertwines the elements of an algebra and its dual, $\mathbb{D}(\Theta_{\mathcal{A}}) = \Theta_{\mathcal{A}^!}$ — an intertwining, not a negation: $\kappa + \kappa^! = 0$ holds on the affine and free lanes and fails already for Virasoro, where the sum is the conductor of Section 10.

5.4. **The primitive kernel.** The Maurer–Cartan element compresses to a primitive kernel: the sum $\mathfrak{K}$ of corolla terms, one for each stable topological type, satisfying the primitive master equation

$$d\mathfrak{K} + \mathfrak{K}\star\mathfrak{K} + \hbar\,\Delta\mathfrak{K} = 0, \tag{5.6}$$

with $\star$ the one-edge sewing and Δ the self-sewing. The first values are structural: the modular characteristic is the trace of the self-sewn binary corolla, $\kappa(\mathcal{A}) = \mathrm{tr}_{\mathrm{cyc}}\,\Delta\,\mathfrak{K}_{0,2}$ — the genus-one tadpole of the two-point function — and the genus-one one-point kernel is $\mathfrak{K}_{1,1} = -P\circ\Delta\,\mathfrak{K}_{0,3}$, the propagator applied to the self-sewn three-point corolla. Genus by genus, the master equation writes each kernel as a polynomial in lower ones: the entire modular structure of a chiral algebra is the free resolution of its operator product expansion by stable curves.

5.5. **Five points, sewn.** The first genuinely global test of the master equation is the five-point sphere, and it passes at chain level. On the formal completion of $\overline{\mathcal{M}}_{0,5}$, with its ten boundary divisors, the five-point correlator of a Koszul algebra extends uniquely across every divisor by the sewing rule

$$\Phi_5\big|_{D_{S,S^c}} = \sum_{a,b} \Phi_3(\xi_S; e_a)\, \eta^{ab}\, \Phi_4(e_b; \xi_{S^c}), \tag{5.7}$$

η^{ab} the inverse Shapovalov form, one rule serving all ten divisors, which the Σ_5-action permutes transitively; the arity-five bar differential is the pullback of the Arnold connection, its square vanishing by the three-term relation together with the infinitesimal braid relations; and the extension is compatible with the genus-one clutching charts through the elliptic trace. The pentagon of $\overline{\mathcal{M}}_{0,5}$ is where associativity of sewing is first a theorem rather than a definition, and the computation here is the base case of the induction of Theorem 5.5.

5.6. **The coderived ambient.** At positive genus the completion becomes part of the structure, and the following theorem locates it exactly.

Theorem 5.6. *For the Virasoro and $\mathcal{W}_N$ algebras the identification of the Batalin–Vilkovisky complex with the bar complex is false at the chain level in the category of bounded direct-sum complexes: the harmonic components of the propagator produce discrepancies $\delta_r = c_r(\mathcal{A})\, m_0^{\lfloor r/2\rfloor -1}$, $r \geq 4$, with $c_4(\mathrm{Vir}_c) = 10/[c(5c+22)] \neq 0$, and no contracting homotopy absorbs them. The discrepancies are multiples of powers of the curvature; their cone is coacyclic. Consequently the identification holds in Positselski's coderived category, equivalently on the pro-object, adic, or weight-completed presentations, which are canonically equivalent; the element $\xi = (e_2, e_3, \dots)$ with $e_k = s^{-1}L_{-k}\mathbf{1} \otimes s^{-1}L_{-k}\mathbf{1}$ lies in the product completion and not in the direct sum: the completion is intrinsic.*

Proof. Hodge-decompose the genus-g propagator into the logarithmic kernel, an exact part, and a harmonic part. The exact part is a chain homotopy, so the whole discrepancy between the two differentials concentrates in the harmonic directions, where degree counting on the Siegel space forces each correction to be a multiple of a power of the central curvature. A cone built from curvature-torsion is resolved by the one-variable Koszul complex of m_0, hence coacyclic; inverting coacyclics — the definition of the coderived category — makes the comparison an equivalence. The three completed presentations are identified by the finite-window stabilisation of Theorem 3.4. □

Theorems 5.1–5.6 say one thing: nilpotence is the logarithm of periodicity. The bar construction is the categorical logarithm — it takes tensor products to direct sums — the cobar construction is the exponential, the Koszul locus is the radius of convergence, κ is the infinitesimal generator of monodromy, and the coderived category is the analytic continuation.

6. The shadow tower and the classification

6.1. **The tower.** Restrict $\Theta_{\mathcal{A}}$ to a one-dimensional primary line L and expand: $\Theta_{\mathcal{A}}|_L = \sum_{r\geq 2} S_r\, x^r$. The coefficients S_r are the *shadow tower* of $\mathcal{A}$; they are the complete set of scalar obstructions to extending the truncations $\Theta^{\leq 2} \to \Theta^{\leq 3} \to \cdots$, and by Theorem 5.3 they are genus-zero data. The first three have independent names and meanings: $S_2 = \kappa$ is the modular characteristic of Theorem 5.1; S_3 is the cubic collision residue; S_4 is the quartic contact term, computed for the Virasoro algebra by the Zamolodchikov norm: the unique quasi-primary at level four is $\Lambda = :TT: -\frac{3}{10}\partial^2 T$ with $\langle \Lambda | \Lambda \rangle = c(5c+22)/10$, and

$$S_3(\mathrm{Vir}_c) = 2 \quad \text{(independent of } c\text{)}, \qquad S_4(\mathrm{Vir}_c) = \frac{10}{c\,(5c+22)}. \tag{6.1}$$

The quartic seed is a Gram computation. At level four the Verma module of Vir_c has the two-dimensional basis $\{L_{-2}^2 v,\, L_{-4}v\}$ modulo derivatives, with Gram matrix $\begin{pmatrix} 5c & 3c \\ 3c & c(c+8)/2 \end{pmatrix}$ of determinant $c^2(5c+22)/2$; the quasi-primary direction is Λ, its norm is the Zamolodchikov value, and S_4 is its inverse. The two zeros of the Gram determinant, $c = 0$ and the Lee–Yang point $c = -22/5$, are the only poles of every shadow coefficient of the family: the Kac determinant at level four governs the arithmetic of the entire tower. The quintic obstruction is the first genuinely nonlinear one, $o^{(5)} = \{S_3, S_4\}_H = 480/[c^2(5c+22)] \neq 0$, and its non-vanishing is the one-line proof that the Virasoro tower is infinite.

Theorem 6.1 (Algebraicity)**.** *Set $H(t) = \sum_{r\geq 2} rS_r t^r$, $\alpha = S_3$, $\Delta = 8\kappa S_4$. The Maurer–Cartan recursion on the line L is equivalent to the single algebraic identity*

$$H(t)^2 \;=\; t^4\, Q(t), \qquad Q(t) \;=\; (2\kappa + 3\alpha t)^2 + 2\Delta\, t^2. \tag{6.2}$$

Three numbers (κ, α, S_4) determine the entire tower: $S_r = \frac{1}{r}\,[t^{\,r-2}]\sqrt{Q(t)}$. The coefficients are rational in the parameters of the family; for the Virasoro algebra the denominators are confined to $c^{\,r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$, and $S_5 = -48/[c^2(5c+22)]$.

Proof. Write $a_n = (n+2)S_{n+2}$. The degree-r component of (5.5) on L is the vanishing of the convolution coefficient $\sum_{i+j=m} a_i a_j$ for $m \geq 3$; isolating the terms with $i = 0$ or $j = 0$ gives $4r\kappa S_r$ against the restricted bracket sum, which is the Taylor recursion for $\sqrt{Q}$; the coefficients of orders $0, 1, 2$ are the definition of Q. ☐

Theorem 6.2 (Depth gap)**.** *Let $r_{\max}$ be the largest r with $S_r \neq 0$. On a single primary line, $r_{\max} \in \{2, 3, \infty\}$: the tower terminates if and only if Q is a perfect square, and a square-root of a non-square quadratic never truncates. Depth four is realised only by a family phenomenon: on the $\beta\gamma_\lambda$ family, $S_3 = 0$ and $S_4 = -5/12$ with $S_{r\geq 5} = 0$ by rank-one rigidity of the* *contact stratum, a two-parameter statement invisible on any single line. The list of depths is complete.*

The discriminant $\Delta = 8\kappa S_4$ decides everything: $\Delta = 0$ forces termination, $\Delta \neq 0$ forces the infinite tower.

6.2. **The shadow connection.** The tower carries its own differential equation. The logarithmic connection $\nabla^{\mathrm{sh}} = d - \frac{Q'(t)}{2Q(t)}\,dt$ is flat with residue $\frac{1}{2}$ at each simple zero of Q and monodromy -1 around it — the Koszul sign realised as monodromy — and the generating function satisfies the Picard–Fuchs equation $2Qf'' + Q'f' - Q''f = 0$ of the genus-zero spectral curve $y^2 = Q(t)$. The tower is therefore a family of periods of a rational curve with four branch points, its large-order behaviour a WKB analysis of the associated Schrödinger operator, and its classical Voros period is π, universally: the semiclassical structure of the obstruction theory is the same for every chiral algebra, and only the position of the branch points — the three seeds — distinguishes one family from another.

6.3. **The classification.**

Theorem 6.3 (Classification)**.** *The chiral algebras of the standard landscape fall into four classes and one critical stratum, according to the shadow tower, and membership is forced by an identifiable mechanism:*

Class	$r_{\max}$	S_3	S_4	mechanism and archetypes
G	2	0	0	*pure double poles: Heisenberg, fermion, lattice*
L	3	$\neq 0$	0	*Jacobi identity kills S_4: affine $\widehat{\mathfrak{g}}_k$, $k \neq -h^\vee$*
C	4	0	$\neq 0$	*quartic contact, family stratum: $\beta\gamma$, bc*
M	∞	$\neq 0$	$\neq 0$	*T is a strong generator:* Vir, $\mathcal{W}_N$, $V^\natural$
FF	—	—	—	*critical level $k = -h^\vee$: $\kappa = 0$, uncurved, non-Koszul*

For class L *the vanishing of S_4 is the Jacobi identity in the form*

$$\textstyle\sum_{(ij|kl)} \kappa_{\mathfrak{g}}([x_i, x_j], [x_k, x_l]) = \langle \mathrm{Jac}, x_4 \rangle = 0,$$

uniform over all simple types, the dual Coxeter number carrying the non-simply-laced cases. Class M *is characterised by self-reference: $r_{\max} = \infty$ if and only if T appears in its own first product, $T \in T_{(1)}T$. The critical stratum is genuinely distinct: at $k = -h^\vee$ the bar complex is uncurved and yet not Koszul, the Feigin–Frenkel centre* $\mathrm{Fun}\big(\mathrm{Op}_{^L\mathfrak{g}}(D)\big)$ *spreading its cohomology off the diagonal* [FF]*. Uncurved and Koszul are independent conditions.*

Proof of the class assignments. Class G: a pure double-pole algebra has scalar residue differential; all products of shadows beyond degree two vanish identically. Class L: the cubic shadow is the structure constant tensor, and the quartic obstruction is the Hessian $\frac{1}{2}\{S_3, S_3\}$, whose value on any test quadruple is a sum of Killing-paired double brackets; the Jacobi identity is exactly its vanishing, as the displayed $\mathfrak{sl}_2$ computation $\frac{2}{k+2} + 0 - \frac{2}{k+2} = 0$ shows in coordinates. Class C: on the $\beta\gamma_\lambda$ family the cubic vanishes by ghost-number parity and the quintic and beyond vanish by rank-one rigidity — the deformation direction is one-dimensional, and every higher transferred product factors through the vanishing bracket of degree two. Class M: with $T \in T_{(1)}T$ the seeds κ, S_3, S_4 are all nonzero, Q has nonzero discriminant, and Theorem 6.2 forces the infinite tower; the quintic value $480/[c^2(5c+22)]$ exhibits the first nonvanishing obstruction explicitly. The critical stratum is Example 2.6 at $k = -h^\vee$ together with the Feigin–Frenkel description of the centre. □

Theorem 6.4 (The bell)**.** *Three integers attached to $\mathcal{A}$ coincide: the shadow depth $r_{\max}$; the A_∞-depth of the minimal model of $\mathcal{A}$ (the largest n with $m_n \neq 0$); and the L_∞-formality level of the genus-zero convolution algebra. A chiral algebra's homotopical complexity is a single number, and class G is precisely the formal class.*

Proof. Homotopy transfer identifies the spectral-sequence differentials d_r with the transferred operations m_{r+1}, so the first nonvanishing higher product and the depth of the obstruction tower coincide; the Kadeishvili tree formula and the stable-graph expansion of the tower run over the same trees with the same propagators, giving the third equality. □

Three separations complete the picture. The classification is a classification of towers, not of central charges: the Moonshine module and the Leech lattice algebra share $c = 24$ and separate as $(\kappa, \text{class}) = (12, \mathsf{M})$ versus $(24, \mathsf{G})$. The class and conductor are independent refinements (Section 10): Virasoro and $\mathcal{W}_3$ are both class M with conductors 13 and 250/3. And three pole-invariants are independent: the maximal generator pole $p_{\max}$, its $d\log$-absorption $k_{\max} = p_{\max} - 1$, and $r_{\max}$ — maximally split by $\beta\gamma$, where they equal $(1, 0, 4)$. Drinfeld–Sokolov reduction is the class escalator $\mathsf{L} \to \mathsf{M}$: gauge symmetry is turned into gravity by quantised Hamiltonian reduction, and this is a construction, not a metaphor.

6.4. **Arithmetic of the Virasoro tower.** The Virasoro tower rewards exact computation. Beyond (6.1) and $S_5 = -48/[c^2(5c+22)]$, the denominators obey the confinement law $c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}$: every pole of every shadow coefficient sits at $c = 0$ or at the Lee–Yang point $c = -22/5$, the two zeros of the level-four Gram determinant $c^2(5c+22)/2$ — the Kac determinant governing the arithmetic of the entire tower from its first nontrivial case. The dual towers satisfy the reciprocity

$$\Delta(c) + \Delta(26-c) \;=\; \frac{6960}{(5c+22)(152-5c)}, \qquad 6960 = 2^4 \cdot 3 \cdot 5 \cdot 29, \tag{6.3}$$

whose two poles are the Lee–Yang points of the dual pair. The branch field of the tower is $\mathbb{Q}\big(\sqrt{-5(5c+22)}\big)$, preserved by the Koszul involution only at $c = 1$ and the self-dual point $c = 13$. Finally, the genus at which a given coefficient first contributes to the free energy is $g_{\min}(S_r) = \lfloor r/2 \rfloor + 1$:

the tower is graded by visibility, and the quartic term is a genus-three phenomenon on the nose.

The closed forms continue: with denominators as stated,

$$S_6 = \frac{80(45c+193)}{3c^3(5c+22)^2}, \qquad S_7 = -\frac{2880(15c+61)}{7c^4(5c+22)^2}, \tag{6.4}$$

and at leading order in $1/c$ the whole tower collapses to a one-parameter family, $S_r = \frac{2}{r}(-3)^{r-4}(2/c)^{r-2} + O(c^{-(r-1)})$, whose Stieltjes transform is a single logarithm: the leading Virasoro tower is the expansion of $-\frac{c^2}{162}\log(1+6t/c)$, one Kummer motive with parameter $-6/c$, iterated against itself. This is the algebraic channel of Section 16 in its most concrete form.

6.5. **Growth and resurgence.** The square root in (6.2) places the branch points of Q in control of the asymptotics: $S_r \sim A\,\rho^r r^{-5/2}\cos(r\theta+\varphi)$ with $\rho = \sqrt{9\alpha^2+2\Delta}/2|\kappa|$. For the Virasoro family $\rho(c)^2 = (180c+872)/[c^2(5c+22)]$; the tower converges for c beyond the largest root $c_* \approx 6.125$ of $5c^3+22c^2-180c-872$, so every unitary minimal model sits in the divergent regime while $c=26$ converges. The Borel transform of the tower is entire; the field-theoretic series it shadows is Gevrey-1 with the Koszul-dual saddle as its Borel singularity. The shadow is the convergent projection of a divergent theory.

7. The landscape in detail

The classification of Theorem 6.3 is only as good as the computations that force it. This section records them, family by family, at chain level.

7.1. **Heisenberg: the atom.** For $\mathcal{H}_k$ every structure of the paper is visible in closed form. The bar chain space in degree n has dimension $p(n-2)$, the partition number, and the differential acts only through the double pole. The degree-two computation $d\big([\alpha|\alpha]\otimes\eta_{12}\big) = k\cdot\mathbf{1}$ exhibits the curvature as the level; the degree-three cancellation is the whole mechanism of Theorem 2.2 in miniature: on $\alpha^{\otimes 4}\otimes\eta_{12}\wedge\eta_{23}\wedge\eta_{34}$ only the three adjacent divisors contribute, non-adjacent pairs dying by $\eta\wedge\eta=0$ or by degree overflow, and d^2 telescopes to $k^2-k^2=0$. The collision residue is $r(z)=k/z$; its path-ordered exponential is the scalar R-matrix $R(z)=e^{k\hbar/z}$; and the Koszul dual is the curved symmetric coalgebra of Example 2.4. The genus-one block is the Fredholm determinant of Section 13; the genus-g block is the string measure of Section 20. Nothing in the general theory is active here, and the general theory suffices in full.

7.2. **Affine algebras: two channels at work.** For $\widehat{\mathfrak{sl}}_2$ at genus two the curvature computation of Theorem 5.1 splits cleanly. The double-pole channel contributes because $\partial_w\omega_j \neq 0$ from genus two on, giving $3k/4$ after the trace $\kappa^{ab}\kappa_{ab}=3$ and the normalisation by $2h^\vee=4$; the simple-pole channel closes a loop through the Killing contraction $\sum_{c,d} f^{ac}_{\ \ d} f^{bc}_{\ \ d} = 2h^\vee\kappa^{ab}$ and contributes $\dim\mathfrak{g}/2 = 3/2$; the total is $\kappa = 3(k+2)/4$, vanishing at the critical level $k = -2$ where the two channels cancel exactly. Nilpotency mixes the channels: $\{d^{(0)}, d^{(2)}\} + (d^{(1)})^2 = 0$ with both summands individually nonzero — a cancellation invisible in the abelian case, and the precise content of Proposition 2.3.

The Serre relations enter as degree-three bar cocycles, and the Jacobi identity kills the quartic shadow: on the test vector (e, f, e, f) the three channels contribute $\frac{2}{k+2} + 0 - \frac{2}{k+2} = 0$. This is class L termination, witnessed rather than asserted. The bar cohomology, of dimension $2n+1$ in degree n, is Chevalley–Eilenberg cohomology of the loop algebra corrected by the Orlik–Solomon factor: for the Witt algebra the mode-algebra count gives $\dim H^1_{\mathrm{CE}} = 2$ where the chiral answer is 1, and the discrepancy is exactly the collapse of non-consecutive collisions: the chiral bar complex is its own object, with its own count.

7.3. **Level one and the unique class transition.** At level one the generic formula $\kappa = \dim \mathfrak{g}\,(k + h^\vee)/2h^\vee$ fails for simply-laced $\mathfrak{g}$: the value it predicts for $\widehat{\mathfrak{sl}}_2$ is $9/4$, the true value is 1. The mechanism is the Frenkel–Kac construction: the root currents are composites of $\operatorname{rk}\mathfrak{g}$ Heisenberg fields, the simple-pole channel dies, the cubic shadow vanishes, and the algebra drops from class L to class G with $\kappa = \operatorname{rk}\mathfrak{g}$. This is the unique crossing between classes on the standard landscape, and it shows κ is an invariant of the algebra, not of a presentation: it must be computed, not read off a formula.

7.4. **Symmetric orbifolds.** The modular characteristic is linear under symmetrisation: $\kappa(\operatorname{Sym}^N \mathcal{A}) = N\kappa(\mathcal{A})$, by four independent mechanisms — the twisted sectors carry positive weight and decouple from the genus-one trace; the logarithm of the Dijkgraaf–Moore–Verlinde–Verlinde product isolates the linear term; the Hecke operator T_N multiplies the vacuum obstruction by N; and the bar complex of the orbifold splits into N copies at genus one. Four proofs of one linearity is the redundancy the subject affords: every scalar statement in this paper is overdetermined.

7.5. **The $\beta\gamma$ system: three invariants split.** The $\beta\gamma$ system of weight λ has $\kappa = 6\lambda^2 - 6\lambda + 1$, the Mumford exponent; its collision residue vanishes, the surviving ordered datum being the contact tensor $\Theta = \beta \otimes \gamma - \gamma \otimes \beta$ with $\Theta^2 = 0$ and $R = \mathrm{id} + 2\pi i\,\Theta$; its Koszul dual is the bc system, the residue pairing exchanging the Weyl and Clifford relations by an explicit four-term cancellation. On the weight family, $S_3 = 0$ and $S_4 = -5/12$ with all higher shadows vanishing — the class-C witness — while on the induced Virasoro line at $c(\lambda)$ the tower is infinite. The triple $(p_{\max}, k_{\max}, r_{\max}) = (1, 0, 4)$ is maximally split: pole order, absorbed pole order, and shadow depth are three independent invariants, and $\beta\gamma$ separates them.

7.6. **Bar dimensions and dual series.** The Hilbert series of the bar cohomologies close into algebraic functions whose branch data is itself an invariant. Writing $h_n = \dim H^n$ on $\mathbb{P}^1$:

algebra	*generating function*	*discriminant*
$\mathcal{H}_k$	$1/(1-x)$-type (partitions shifted)	1
$\beta\gamma$	$\sqrt{(1+x)/(1-3x)}$	$(1-3x)(1+x)$
$\widehat{\mathfrak{sl}}_2$	$x(3-x)/(1-x)^2$	$(1-3x)(1+x)$
Vir_c	$4x/\big(1-x+\sqrt{1-2x-3x^2}\big)^2$	$(1-3x)(1+x)$
$\widehat{\mathfrak{sl}}_3$	$4x(2-13x-2x^2)/[(1-8x)(1-3x-x^2)]$	$(1-8x)(1-3x-x^2)$

The rank-one families share the branch divisor $(1-3x)(1+x)$ with dominant growth $3^n = \dim\mathfrak{sl}_2$; the rank-two family grows as $8^n = \dim\mathfrak{sl}_3$ with the golden-ratio-type factor $1-3x-x^2$. Growth is governed by the dimension of the underlying Lie algebra; the subdominant factor is a finer invariant, stable under Drinfeld–Sokolov reduction at the level of branch charts. The Virasoro dimensions are first differences of Motzkin numbers; the entropy $h = \log(\text{growth rate})$ of the completed dual ascends strictly along the $\mathcal{W}$-tower, from $h(\mathcal{W}_2) \approx 0.567$ to the MacMahon value $h(\mathcal{W}_\infty) \approx 0.872$: each additional generator weight adds strictly positive completion entropy.

7.7. **The boundary of the landscape.** The Ising algebra is class M with an asymptotic, Borel-summable tower ($S_4 = 40/49$, growth $\rho \approx 12.5$): a rational, solvable conformal field theory whose shadow expansion diverges — divergence of the tower is a statement about the expansion, not about the theory. The Moonshine module has $\kappa = c/2 = 12$, the double-pole channel carrying everything at weight one, and $\Delta = 20/71 \neq 0$ places it in class M; the Leech lattice algebra at the same central charge is class G with $\kappa = 24$. The logarithmic triplet algebras $\mathcal{W}(p)$ bound the landscape on the non-semisimple side: C_2-cofinite with $2p$ simple modules and provably outside finite free strong generation, they carry unbounded Massey structure where the standard classes carry towers, and the classification of this paper stops at their doorstep by construction — the class axioms presume the semisimple vacuum geometry the triplets are built to violate. The critical-level algebra $V_{-h^\vee}(\mathfrak{g})$ is uncurved, non-Koszul, with trivialised monodromy and tripled degree-two cohomology: every boundary phenomenon of the theory, in one example.

7.8. **Kinematics and dynamics.** The bar theory of an algebra splits into a kinematic and a dynamic layer. The vacuum character alone fixes the kinematics: the growth of the chain spaces, the entropy of the completed dual, the radius of the Koszul series through the functional equation (3.1). The operator product fixes the dynamics: the differential, the tower, the class. Two algebras with the same character and different products — the Leech and Moonshine theories at $c = 24$ — share every kinematic invariant and separate at the first dynamic one, which is the cleanest justification for the central place the tower occupies in this paper.

7.9. **The census.** The standard landscape, in the counting of this paper, comprises twenty-one families: five Gaussian, two Lie-type, four contact, and ten gravitational, with the critical stratum transversal to all. The three pole-invariants, the shadow class, the modular characteristic, and the conductor are tabulated in Appendix A; every entry is the output of the two-channel computation of Theorem 5.1 and the recursion of Theorem 6.1, and the table is closed under Koszul duality, level reflection, and Drinfeld–Sokolov transport.

8. Drinfeld–Sokolov and the $\mathcal{W}$ quantum group

8.1. **The escalator.** Quantised Drinfeld–Sokolov reduction along a principal nilpotent maps class L to class M: it manufactures gravity from gauge theory. At chain level the mechanism is homotopy transfer: the reduction contracts the bar complex along the BRST differential, and the transferred A_∞ operations are sums over planted binary trees — Catalan-many at each arity — whose internal edges carry the Drinfeld–Sokolov homotopy. The transfer sends the simple-pole r-matrix of the affine algebra to the third-order kernel $r(z) = (c/2)z^{-3} + 2Tz^{-1}$ of the Virasoro algebra; it escalates the pole by two, and with it the depth from 3 to ∞. The coproduct is transferred to its primitive part: a ghost-number argument annihilates every group-like correction, which is the coalgebraic statement behind the statement of Section 20 that gravity braids, and braiding is all it does.

For $\mathcal{W}_3$ the transferred data are explicit: the generator W of weight three has $W_{(5)}W = c/3$ and $W_{(1)}W = \frac{3}{10}\partial^2 T + \frac{16}{22+5c}\Lambda$, with Λ the quasi-primary of (6.1); the two-channel decomposition $\kappa(\mathcal{W}_3) = \frac{c}{2} + \frac{c}{3} = \frac{5c}{6}$ weights the generators by their Eulerian factors; the central-charge conductor is $c + c^! = 100$ with self-dual point $c = 50$. The non-principal reductions test the limits: for the Bershadsky–Polyakov algebra $\mathcal{W}^k(\mathfrak{sl}_3, f_{\min})$, with $c(k) = -(2k+3)(3k+1)/(k+3)$, the reflection $k \mapsto -k-6$ gives the exact conductor $c(k) + c(-k-6) = 50$; the algebra is the first with mixed shadow classes — class G on its current line, class M on its stress line — and its modular characteristic is the genus-one curvature of a non-principal reduction, the next computation of the theory. For the $(3,2)$ nilpotent in $\mathfrak{sl}_5$ the BRST differential contains the cubic ghost term $c_{12}c_{23}b_{13}$: the dual object is an inhomogeneous Koszul complex, and it governs the transport of duality along non-hook reductions.

For hook-type nilpotents the duality transports along the reduction: granting the compatibility of the reduction with the bar construction edge by edge, the dual of $\mathcal{W}^k(\mathfrak{sl}_N, f_\lambda)$ is $\mathcal{W}^{-k-2N}(\mathfrak{sl}_N, f_{\lambda^t})$, the transpose partition at the reflected level — Koszul duality exchanging a nilpotent orbit with its transpose, which is the affine shadow of the duality of Barbasch and Vogan [BV].

8.2. **The quantum group of $\mathcal{W}_{1+\infty}$.** The ordered theory of Section 4 assigns to the $\mathcal{W}_{1+\infty}$ family a genuine quantum-group structure, and its failure to be Hopf is a theorem. With parameter Ψ and the scalar Maulik–Okounkov R-matrix $g(z) = \prod_i (z-h_i)/(z+h_i)$ [MO], the Drinfeld coproduct is spectral,

$$\Delta_z\big(T(u)\big) = T(u)\otimes T(u-z), \qquad \Delta_z(T) = T\otimes 1 + 1\otimes T + \frac{\Psi-1}{\Psi}\, J\otimes J + z\,(1\otimes J) + \cdots, \tag{8.1}$$

and the shift $u \mapsto u-z$ is forced by the conformal anomaly: the obstruction class of Theorem 20.1 is exactly what a z-independent coproduct would have to kill. The Miura cross-term coefficient $(\Psi-1)/\Psi$ is independent of the spin — verified through spin six — and reappears, with the same value, as the obstruction cocycle $[Q_\alpha, \Delta_z^{\mathfrak{h}}] = \frac{\Psi-1}{\Psi}\, e^{\alpha\cdot\varphi} \otimes e^{\alpha\cdot\varphi}(1+O(z))$ to descending the free-field coproduct through the screening: the same number measures the failure from both sides. An antipode exists at the level of an involution, $S(J) = -J$, $S(T) = -T + \frac{\Psi-1}{\Psi} :JJ:$, $S^2 = \mathrm{id}$, but it is a morphism of vertex algebras only at $\Psi \in \{1, 2\}$, and the Hopf axiom fails at every Ψ with residual $-(\Psi-1)/z^2 + zJ$: the object is a chiral bialgebra and provably not a Hopf algebra. For $\mathfrak{gl}_N$ the quantum determinant is central only for the decreasing column ordering with spectral shifts $u-(i-1)\Psi$; the ordering is rigid, the shift pattern forced.

8.3. **Genus one for the ordered theory.** At genus one the ordered bar complex of the affine algebra carries the elliptic Knizhnik–Zamolodchikov–Bernard connection; its B-cycle monodromy is q^{2H} with $q = e^{\pi i/(k+h^\vee)}$, and the elliptic collision kernel decomposes through the Weierstrass functions, $r^{(1)} = c_0\,\zeta_\tau + \sum_{n\geq 1} \frac{(-1)^{n-1}}{n!} c_n\,\wp^{(n-1)}$, with B-cycle defect $-2\pi i\, c_0$. The dichotomy of the landscape reappears analytically: the curvature and the braiding are entangled exactly when $c_0 \neq 0$, that is for the affine, Virasoro and $\mathcal{W}$ families, and decoupled for the free families. The Belavin elliptic r-matrices, the Felder dynamical equation, and — at genus two — a doubly-dynamical structure over the Siegel space with the prime-form kernel, all arise as specialisations; at genus two the correct object is a Hall–Drinfeld double rather than a Yangian, the Manin-triple signature $(2,1)$ admitting no Lagrangian splitting.

8.4. **Seven faces of the r-matrix.** The binary collision residue (4.2) admits seven presentations in the literature of integrable systems: as the binary component of the universal twisting morphism; as a line-operator r-matrix; as the Laplace transform of a classical λ-bracket; as the symbol of commuting sphere Hamiltonians; as Drinfeld's classical r-matrix; as the kernel of the Sklyanin bracket; and as the kernel of the Gaudin Hamiltonians, which arise from it by taking residues, $H_i = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$, with the Knizhnik–Zamolodchikov coupling $1/(k+h^\vee)$. At genus zero the seven faces coincide on the binary slice for a reason of Grothendieck–Teichmüller type: the group GRT_1 has no generator in weight two, so no gauge separates them. At genus one the quasi-period datum activates the higher weights and the seven elliptic faces separate; the set of presentations becomes a torsor over $\mathrm{GRT}_1(\mathbb{Q})$, which is the same torsor that indexes splittings of the averaging map in Section 4. That the ambiguity of quantisation and the ambiguity of presentation are one group is forced; it is the associator doing its one job in two costumes.

9. The moduli of Koszulness and the fingerprint

9.1. **Koszulness as a moduli problem.** Koszulness is a space. For fixed $\mathcal{A}$, consider the functor of points whose values are triples: an associator Φ; a Maurer–Cartan deformation ξ of the tower; and a trivialisation of the cone of the quadratic comparison $q_{\mathcal{A}}$. At each finite Postnikov stage with framing this is an open subfunctor of an affine Maurer–Cartan scheme, cut out by Fitting ideals; the associator coordinate makes the whole object a torsor over the Grothendieck–Teichmüller group, and Koszulness at one associator implies it at every other. The moduli perspective earns its keep twice. First, it organises the fourteen known recognition tests — among them the PBW collapse, the A_∞-formality, the monadicity criterion, the diagonal concentration of factorization homology, the mixed-Hodge purity test at the de Rham–Betti associator, and the Lagrangian transversality test — as charts of one atlas, each valid at its own associator base point: ten of the tests are unconditional biconditionals, three are one-directional, and the remainder carry hypotheses. Second, it locates the failures of Theorem 3.3 as boundary strata: the Virasoro chart closes non-circularly through the Feigin–Fuchs resolution, the Kac determinant, and Hodge purity, while the admissible quotients sit outside every chart, their Cartan obstructions visible from all of them.

9.2. **The fingerprint.** The classification data assemble into the fingerprint $\varphi(\mathcal{A}) = (p_{\max}, r_{\max}, \chi, n_{\text{strong}}, \text{coset})$: maximal generator pole, shadow depth, character, number of strong generators, coset datum. On the standard landscape the fingerprint together with the Koszul dual determines the bar coalgebra up to isomorphism. The semion and its conjugate share every scalar and differ by a twist: the twist is the datum beyond the fingerprint, and the reconstruction clause carries a residual-freeness hypothesis for exactly this reason. The coarse projection of the fingerprint is the class of Theorem 6.3, and each projection has a computable fibre: Virasoro and $\mathcal{W}_3$ share a class and carry conductors 13 and 250/3. What survives every projection is κ; what survives none of them is the braiding.

10. Complementarity and the conductor

10.1. **The splitting.** Let $\mathcal{Z}(\mathcal{A})$ be the centre local system on $\overline{\mathcal{M}}_g$ attached to a chirally Koszul $\mathcal{A}$ with Koszul dual $\mathcal{A}^!$, and let $\mathbf{C}_g(\mathcal{A}) = R\Gamma\big(\overline{\mathcal{M}}_g, \mathcal{Z}(\mathcal{A})\big)$.

Theorem 10.1 (Complementarity)**.** *The Verdier involution σ acts on $\mathbf{C}_g(\mathcal{A})$, and its eigenspace decomposition splits the ambient cohomology into the contributions of the two dual algebras,*

$$\mathbf{C}_g(\mathcal{A}) \;\simeq\; \mathbf{Q}_g(\mathcal{A}) \,\oplus\, \mathbf{Q}_g(\mathcal{A}^!) : \tag{10.1}$$

what one algebra counts as deformations, its dual counts as obstructions. If in addition the pairing induced by Verdier duality is perfect and anti-invariant, the two summands are complementary Lagrangians for a symplectic structure of degree $-(3g-3+n)$, matching the dimension of moduli. The scalar trace of the splitting is the conductor

$$\varkappa(\mathcal{A}) \;=\; \kappa(\mathcal{A}) + \kappa(\mathcal{A}^!), \tag{10.2}$$

and its unconditional values on the standard landscape are:

$$\varkappa = 0 \ \ (\textit{Heisenberg, free fields, lattices, affine under } k \mapsto -k-2h^\vee), \qquad \varkappa(\mathrm{Vir}) = 13. \tag{10.3}$$

Proof of the splitting. The involution exists on the represented strict-flat locus: Verdier duality on the Ran space composed with the centre comparison defines σ with $\sigma^2 = \mathrm{id}$, and in characteristic zero the idempotents $\frac{1}{2}(1 \pm \sigma)$ split any complex on which an involution acts. The identification of the eigenspaces is the content: the bar object of $\mathcal{A}$ is the j_*-extension across the collision divisors and lands in the $+1$ eigenspace, while the bar object of $\mathcal{A}^!$ is the $j_!$-extension, and the Kodaira–Spencer anti-commutation $\mathbb{D} \circ \nabla_\kappa = -\nabla_\kappa \circ \mathbb{D}$ — contravariance of duality reversing infinitesimal actions — places it in the -1 eigenspace. Isotropy of each summand and perfectness of the cross-pairing follow from anti-invariance of the Verdier pairing when that pairing is perfect, which on the standard landscape is verified through the PBW filtration: the associated graded sees only the genus-zero differential, Koszulness concentrates it in one row, and cohomology-and-base-change transports perfectness across moduli. The scalar values are Theorem 5.1 plus arithmetic: $\kappa(V_k) + \kappa(V_{-k-2h^\vee}) = 0$ term by term, and $\frac{c}{2} + \frac{26-c}{2} = 13$. $\square$

Theorem 10.2 (Continuation of Theorem 10.1)**.** *The Virasoro value restates the arithmetic identity $c + c^! = 26$, with self-dual point $c = 13$. Under the hypothesis that Drinfeld–Sokolov reduction commutes with the bar construction, the principal $\mathcal{W}_N$ family carries $\varkappa = (H_N - 1)\big(4N^3 - 2N - 2\big)$, in particular $\varkappa(\mathcal{W}_3) = 250/3$ with self-dual point $c = 50$; the central-charge conductors $c + c^! = 2\dim\mathfrak{g}$, 26, 100, and $4N^3 - 2N - 2$ are exact identities of rational functions of the level.*

The two scalar conductors are related by the anomaly ratio: with $K = c + c^!$ and $\varrho = \kappa/c = \sum_i 1/(m_i+1)$ over the exponents, $\varkappa = \varrho\, K$. The ghost form of K explains its values: $K = -c_{\mathrm{ghost}}$ of the canonical BRST complex, and $\sum_{j=2}^{N} 2(6j^2 - 6j + 1) = 4N^3 - 2N - 2$ recovers 26 at $N = 2$. That the critical dimension of the bosonic string appears here as the Virasoro conductor — twice the self-dual point — is the same cancellation, met from the algebraic side.

Conjecture 10.3. $\mathrm{Vir}_c^! \simeq \mathrm{Vir}_{26-c}$ *as chiral algebras. The scalar shadow of this statement, $\varkappa = 13$, is Theorem 10.1; the fixed point of the reflection is the self-dual algebra at $c = 13$.*

Conjecture 10.4. *Let $\widetilde{H}(K3, \mathbb{Z}) = U^4 \oplus E_8(-1)^2$ be the Mukai lattice, of signature $(4, 20)$, so that $2c_+ = 8$. The chiral algebra attached to a $K3$ surface by the two-stage specialisation functor carries conductor $\varkappa = 8$, and the quantisation parameter obeys $\hbar^2\varkappa = -1$, that is $\hbar^2 = -\frac{1}{8}$. The lattice identity $2c_+ = 8$ is a theorem; the conjecture is the recognition of the chiral source.*

For the Bershadsky–Polyakov algebra the central-charge conductor $c(k) + c(-k-6) = 50$ is an identity of rational functions; its modular characteristic is the genus-one curvature of the minimal reduction, and through the ratio law it carries the predicted conductor $\varkappa = 25/3$.

10.2. **The quantisation law.** The conductor fixes the quantum parameter. On the lattice side the identity is arithmetic: for the reflective Borcherds data attached to the Monster, $K3$ and Fake Monster lattices the pairs $(\varkappa, \hbar^2)$ are $(2, -\frac{1}{2})$, $(8, -\frac{1}{8})$ and $(50, -\frac{1}{50})$, one law,

$$\hbar^2\,\varkappa \;=\; -1: \tag{10.4}$$

the square of the quantisation parameter is minus the inverse conductor. On the chiral side the law is Conjecture 10.4 at the $K3$ row and a theorem of lattice arithmetic everywhere it has been evaluated; it is the cleanest statement of the principle that the deformation direction and the duality pairing are inverse structures.

10.3. **Self-dual points.** The fixed locus of the Koszul involution is a subvariety of the space of theories, and it is where the theory is most symmetric. The self-dual points are the half-conductors: $c = 13$ for the Virasoro family, $c = 50$ for $\mathcal{W}_3$, $c = 2N^3 - N - 1$ for $\mathcal{W}_N$; among these the Virasoro point is distinguished as the one attained at real central charge — the $\mathcal{W}_3$ midpoint is reached only at $k = -3 \pm i$, so the Virasoro family alone carries its self-dual theory on the physical line. On the fixed locus the complementarity splitting balances, $\dim \mathbf{Q}_g(\mathcal{A}) = \dim \mathbf{Q}_g(\mathcal{A}^!) = \frac{1}{2}\dim H^*$, and the conductor identity becomes the equivariant signature of the involution. The three involutions of the theory — Verdier, the level reflection, and the braiding inversion $q \mapsto q^{-1}$ — coincide on this locus and only here; at $c = 13$ all three fix the same algebra.

11. The genus expansion

11.1. **The scalar tower.** The genus expansion of $\mathcal{A}$ is the sequence of traces of $\Theta_{\mathcal{A}}$ against the fundamental classes of $\overline{\mathcal{M}}_g$. Its first term is unconditional; the higher terms factor through the Hodge bundle.

Theorem 11.1. *The genus-one trace is $\mathrm{tr}_1\,\mathrm{Obs}^{\mathrm{def}}_{1,1}(\mathcal{A}) = \kappa(\mathcal{A})\,\lambda_1$ in $H^2(\overline{\mathcal{M}}_{1,1})$; equivalently $F_1 = \kappa/24$. Under the K-theoretic comparison of the deformation* *class with $\kappa\,\lambda_{-1}(\mathbb{E}_g)$, the Hodge component at genus g is $(-1)^g\kappa\,\lambda_g$, and for algebras of uniform generator weight the scalar free energies are*

(11.1)
$$F_g^{\mathrm{sc}} = \kappa\,\lambda_g^{\mathrm{FP}}, \qquad \lambda_g^{\mathrm{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}}\,\frac{|B_{2g}|}{(2g)!}, \qquad \sum_{g\geq 1} F_g^{\mathrm{sc}}\,x^{2g} = \kappa\Big(\frac{x/2}{\sin(x/2)}-1\Big),$$

by Mumford's relation and the Faber–Pandharipande evaluation [Mum, FP]*: the generating function is the Wick-rotated $\widehat{A}$-genus, with radius of convergence 2π and universal ratio $F_2/F_1 = 7/240$. The first six values of λ_g^{FP} are*

$$\tfrac{1}{24},\quad \tfrac{7}{5760},\quad \tfrac{31}{967680},\quad \tfrac{127}{154828800},\quad \tfrac{73}{3503554560},\quad \tfrac{1414477}{2678117105664000}.$$

For the E_8 lattice algebra $F_2 = 7/720$, verified independently by the bar complex, by the Siegel–Weil identity $\Theta^{(2)}_{E_8} = E_4^{(2)}$, and by direct enumeration of the 30240 pairs of roots.

The proof of Theorem 11.1 runs along the chain Quillen anomaly, Grothendieck–Riemann–Roch, Faber–Pandharipande: the curvature of the determinant line is $-2\pi i\,c_1(\mathbb{E})$; the Chern character of the Hodge bundle is $g+\sum_{j\geq 1}\frac{B_{2j}}{(2j)!}\,\kappa_{2j-1}$ on the universal curve; the ψ-integrals $\int_{\overline{\mathcal{M}}_{g,1}}\psi^{2g-2}\lambda_g$ evaluate the numbers; and Mumford's identity $\lambda_g^2 = 0$ resums the even Todd series to the sine. Independent confirmations come from the family index theorem and, for the free families, from the operator theory of Section 13.

11.2. **The multi-weight correction.**

Theorem 11.2. *The scalar formula* (11.1) *is exact only on the uniform-weight lane, and the failure beyond is explicit. For $\mathcal{W}_3$ at genus two, the sum over the seven stable graphs of $\overline{\mathcal{M}}_2$, with channel assignments weighted by the two-field propagators $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$ and the $\mathbb{Z}/2$ selection rule killing odd-W vertices, gives*

$$F_2(\mathcal{W}_3) = \frac{7\kappa}{5760} + \delta F_2^{\mathrm{cross}}, \qquad \delta F_2^{\mathrm{cross}} = \frac{3}{c}+\frac{9}{2c}+\frac{1}{16}+\frac{21}{4c} = \frac{c+204}{16\,c}, \tag{11.2}$$

the four terms being the sunset, theta, bridge-loop and barbell graphs — the remaining three of the seven stable graphs of $\overline{\mathcal{M}}_2$ carry an odd number of W-legs at some vertex and vanish by the $\mathbb{Z}/2$ symmetry — and the bridge-loop contribution $1/16$ is a topological residue independent of c. This enumeration is the proof: seven graphs, three killed by parity, four evaluated by the propagators $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$ against the structure constants of $\mathcal{W}_3$: the sum is $\frac{3}{c}+\frac{9}{2c}+\frac{21}{4c} = \frac{51}{4c}$, and $\frac{51}{4c}+\frac{1}{16} = \frac{c+204}{16c}$. The corrections grow with genus faster than the scalar part — the ratios of cross to scalar at $g = 2, 3, 4$ are $O(1)$, $42/31$, $9184/381$ — so the $\widehat{A}$-tower is the leading term, not the sum, of the free energy of an interacting multi-weight algebra. Free-field algebras are the exact multi-weight exception: all their cross-channels vanish identically.

The corrections themselves obey a universal law: for the principal $\mathcal{W}_N$ family the gravitational part of the genus-two correction is

$$\delta F_2^{\mathrm{grav}}(\mathcal{W}_N) = \frac{(N-2)(N+3)}{96} + \frac{(N-2)(3N^3+14N^2+22N+33)}{24\,c}, \tag{11.3}$$

vanishing at $N=2$ as it must, exact at $N=3$, and growing as $N^2/96$ in the planar limit — the 't Hooft scaling emerging from the stable-graph combinatorics alone.

The two theorems together delimit what a scalar can know. The genus tower of any chiral algebra begins with a universal multiple of the $\widehat{A}$-genus; the deviation from universality is a graph calculus with its own arithmetic — the $\mathcal{W}_4$ corrections leave the field $\mathbb{Q}(c)$ — and it, not the scalar, dominates asymptotically.

11.3. **Resurgence of the scalar tower.** The generating function (11.1) is meromorphic with simple poles at $x = 2\pi n$, and its resurgent structure is completely explicit: the large-genus asymptotics $F_g \sim 2\kappa\,(2\pi)^{-2g}$ are saturated by the first pole; the instanton action is $(2\pi)^2$, universal across the landscape; the Stokes constants are $(-1)^n\,4\pi^2 n\,\kappa\,i$; and the Borel transform is entire, so the tower is Borel-summable with optimal truncation at $g \approx \pi/\rho$. Set against the $(2g)!$ growth of the string perturbation series this is the cleanest formulation of the paper's analytic moral: the modular trace projects a factorially divergent theory onto a convergent shadow, and the projection loses exactly the non-perturbative sectors, whose location it nevertheless records in its Stokes data. The Gaussian matching completes the picture: on the scalar lane the free energies coincide with those of the Gaussian matrix model at $N^2 = \kappa$, by Harer–Zagier [HZ]; the shadow of any chiral algebra is a Gaussian theory with coupling κ, and everything beyond the shadow — the graph calculus of Theorem 11.2 — is precisely what distinguishes an interacting algebra from its Gaussian projection.

12. Gaudin, Bethe, and the spectral chain

The integrable structure of a chiral algebra is the ordered theory read spectrally, and it descends from $\Theta_{\mathcal{A}}$ along one chain:

$$\Theta_{\mathcal{A}} \longrightarrow r(z) \longrightarrow R(z) \longrightarrow T(u) \longrightarrow Q(u): \tag{12.1}$$

the Maurer–Cartan element projects to the classical r-matrix, whose path-ordered exponential is the quantum R-matrix, whose trace is the transfer matrix, whose Baxter operator carries the spectrum.

The first arrow is Theorem 5.5 at degree two. The second is the Kohno monodromy of the shadow connection: flatness of $\nabla = d - \sum r_{ij}(z_i - z_j)\,dz_{ij}$ is the classical Yang–Baxter equation, which is the Arnold relation (1.2) contracted with the Casimir; the quantum Yang–Baxter equation for the monodromy follows by path-ordering. The third arrow is the RTT trace, and commutativity of transfer matrices is the geometric statement that disjoint collision cycles commute. The fourth is representation theory on the Baxter locus: the singular vector $w_\lambda = \lambda(v_- \otimes v_\lambda) - v_+ \otimes f v_\lambda$ is a highest-weight vector exactly at $b = a - (\lambda+1)/2$, and the TQ-relation $\Lambda Q = aQ(u-i) + dQ(u+i)$ follows from the resulting short exact sequence of prefundamentals.

Bethe's equations close the circle: the stationarity of the shadow Yang–Yang potential reproduces $\prod_{j\neq i} \frac{u_i - u_j + i}{u_i - u_j - i} = \big(\frac{u_i + i/2}{u_i - i/2}\big)^L$, with energies $E = \frac{L}{4} - \frac{1}{2}\sum_a (u_a^2 + \frac{1}{4})^{-1}$, and the Bethe vectors $B(u_1)\cdots B(u_M)|\Omega\rangle$ live in the ordered tensor product: the ordered bar complex is the chain-level refinement of the algebraic Bethe ansatz, the symmetric shadow retaining the spectrum (Λ, Q) and the ordered kernel the wavefunctions. The Gaudin Hamiltonians $H_i = \sum_{j\neq i} \Omega_{ij}/(z_i - z_j)$ are the residues of the second arrow, with the Knizhnik–Zamolodchikov coupling $1/(k+h^\vee)$; their diagonalisation is the semiclassical face of the same chain.

13. Conformal blocks, Verlinde numbers, and the analytic layer

13.1. **Blocks from the pointed bar complex.** The zeroth cohomology of the bar complex with module insertions computes derived coinvariants; the comparison with the sheaf of conformal blocks of Tsuchiya–Ueno–Yamada [TUY] holds given the integrable truncation, the flat connection, and the sewing and determinant-line data. On the integrable quotient of $\widehat{\mathfrak{sl}}_2$ at level k the identification is complete enough to recover the Verlinde formula [Ver] from chiral homology: with $n = k+2$,

$$Z_g(k) = \sum_{j=1}^{n-1} S_{0j}^{2-2g}, \qquad Z_g = \frac{n^{g-1}}{2^{g-1}} \sum_{j=1}^{n-1} \csc^{2g-2}\Big(\frac{\pi j}{n}\Big), \tag{13.1}$$

a polynomial in n of degree $3g-3$; the closed forms begin $Z_2 = n(n^2-1)/6$ and $Z_3 = n^2(n^2-1)(n^2+11)/180$, and the leading coefficient is $\zeta(2g-2)/(2^{g-2}\pi^{2g-2})$, which is the source of the Bernoulli numerators of Theorem 16.1. The handle operator is S_{0j}^{-2}; the genus-one count $Z_1 = k+1$ is obtained three independent ways — the S-matrix, the Zhu algebra [Zhu] $U(\mathfrak{sl}_2)/(e^{k+1}, f^{k+1})$, and the degree-zero ordered centre. At level one the interacting sewing kernel is diagonalisable in closed form, with eigenvalues $\{q, q^2, q^2\}$ on the three-state block — the first fusion computation of the analytic layer beyond the free families.

13.2. **Characters from the bar resolution.** The classical character formulas are bar cohomology. Filtering the bar complex of $\widehat{\mathfrak{g}}_k$ by the Weyl group, the first page is a sum of Verma characters over affine Weyl chambers, the differential is composed of the singular-vector embeddings located by the

Kac determinant, and the alternating cancellation is the Bernstein–Gelfand–Gelfand resolution run inside the bar complex: the Weyl–Kac formula

$$\operatorname{ch} L(\lambda) = \frac{\sum_w \varepsilon(w)\, e^{w(\lambda+\rho)-\rho}}{\prod_{\alpha>0}\big(1-e^{-\alpha}\big)^{\operatorname{mult}\alpha}} \tag{13.2}$$

drops out of the collapse, and the denominator identities of the free theories are the statement that their bar complexes are resolutions. At admissible fractional level the same mechanism runs with screening operators as the differentials: the Wakimoto free-field realisation [Wak] resolves the vacuum module away from the Kac–Kazhdan hyperplanes, its BRST cohomology concentrating in one degree [Ara], and the Kac–Wakimoto characters collapse, by the Kac–Wakimoto theorem [KW], to a two-term Weyl sum. Free-field realisations are, in this paper's terms, chains of quasi-isomorphisms between bar complexes; the screening charges are the connecting differentials; and the classical character theory of affine algebras is the shadow of Theorem 3.2 on characters.

13.3. **The analytic layer.** The algebraic sewing of Theorem 5.5 acquires an analytic completion under growth hypotheses, and for the free families the completion converges.

Theorem 13.1. *Suppose the graded dimensions of $\mathcal{A}$ grow subexponentially, $\dim \mathcal{A}_n \le Ce^{\alpha\sqrt{n}}$, and the structure constants polynomially. Then the q-damped pair-of-pants operator is Hilbert–Schmidt for every $0<q<1$, and every closed stable amplitude with at least two sewn channels is trace class. For the rank-r Heisenberg algebra the one-particle sewing operator T_q has eigenvalues q^n, trace norm $q/(1-q)$, and Fredholm determinant*

$$\det(1-T_q) \;=\; q^{-1/24}\,\eta(q), \tag{13.3}$$

so the genus-one block is $\eta(\tau)^{-r}$ by second quantisation, and the genus-g block is the string measure of Section 20. For lattice algebras the block acquires the factor $\Theta_\Lambda(\Omega)$, and at genus two the theta factor separates isospectral lattices beyond the reach of genus one.

Proof. The growth hypotheses bound the matrix elements of the damped pair-of-pants by a summable Gaussian in the weights, giving the Hilbert–Schmidt property; a closed amplitude with two sewn channels is a product of two Hilbert–Schmidt operators, hence trace class. For the Heisenberg algebra, Wick's theorem reduces the second quantisation to the one-particle Bergman space, where the sewing operator is multiplication by q^n and the determinant is the Euler product of η. ☐

The Dedekind eta function is an output here: it is the Fredholm determinant of the sewing operator. This is the sense in which the analytic theory is a layer over the same graph sums, and not a second theory.

13.4. **The critical level.** The boundary stratum FF deserves its own statement. At $k = -h^\vee$: the curvature vanishes ($\kappa = 0$), so the bar complex is uncurved at every genus; the monodromy trivialises, $e^{2\pi i/(k+h^\vee)}$ degenerating; Koszulness fails, the cohomology spreading off the diagonal; and the degree-zero bar cohomology is the Feigin–Frenkel centre, the algebra of functions on opers for the Langlands dual group [FF]. Under the comparison of the bar complex with the de Rham complex of the space of opers on the formal disc, the higher cohomology is $\Omega^n(\mathrm{Op}_{^L\mathfrak{g}}(D))$ — a statement on the formal disc, at the threshold where the localisation theory of Frenkel–Gaitsgory begins. The level reflection $k \mapsto -k - 2h^\vee$ is the unique affine involution fixing the critical level and compatible with κ; on quantum parameters it is $q \mapsto q^{-1}$; and its fixed stratum is where the Langlands-dual geometry replaces the finite closed sector of Theorem 15.4. At rank one the critical centre is periodic: the formal-disc Hochschild cohomology of $V_{-2}(\mathfrak{sl}_2)$ is the free graded-commutative algebra $\Lambda(P) \otimes \mathbb{C}[\Theta]$ with $\deg P = 3$, $\deg \Theta = 4$ — four-periodic, the algebraic germ of the modularity that the paramodular forms of Section 19 exhibit globally. Everything singular about the critical level is the statement that flatness and duality degenerate together.

14. The genus-two kernel

Genus two is where the modular theory first outruns every genus-one structure, and each of its new phenomena is computable.

The propagator is built from the Szegő kernel $S_2(z, w) = d \log E(z, w)$ with quasi-periodicity $S_2(z + B_\alpha, w) = S_2(z, w) - 2\pi i\, \omega_\alpha(z)$: the monodromy defect, scalar at genus one, is function-valued at genus two, and through it the level enters the curvature directly (Theorem 5.1). The flat connection acquires three modular directions, one for each entry of the period matrix, with the genus-two heat equation $\partial_{\Omega_{\alpha\beta}}\theta = (2\pi i)^{-1}\partial_{z_\alpha}\partial_{z_\beta}\theta$ coupling them; the twisted local system on the configuration space of a genus-two curve has $\dim H^1 = 12$ at generic monodromy, by the Euler-characteristic count $\chi = r(2 - 2g - s)$. The quantum object attached to the two-holed geometry is a Hall–Drinfeld double: the Manin triple has signature $(2, 1)$, and the two-dimensional positive part against the one-dimensional negative part replaces the Lagrangian splitting of the Yangian — genus two quantises a double, and the doubling is forced by the signature.

On the automorphic side the genus-two kernel reaches what genus one provably organises differently: the Siegel theta series separates $E_8 \oplus E_8$ from D_{16}^+; the two-variable sewing kernel carries an off-diagonal Schur complement whose spectral decomposition reaches central values of twisted L-functions through the Igusa form $\Phi_{10} = \Delta_5^2$; and the second-quantised $K3$ elliptic genus $1/\Phi_{10}$ sits at the boundary of the classification, where the Borcherds algebras of Section 19 begin. The genus-two kernel is the smallest window through which the arithmetic of Section 16, the automorphic boundary, and the gravitational classes are all visible at once.

15. The bulk

15.1. **The derived centre as the closed sector.** The bar complex of Section 2 lives on configurations in a complex curve crossed with an ordered real line: the holomorphic factor carries the residues, the topological factor the deconcatenation. This is the geometry of a three-dimensional holomorphic–topological theory on $X \times \mathbb{R}$, encoded operadically by the two-coloured Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$, whose closed colour is $\mathrm{FM}_k(\mathbb{C})$, whose mixed part is $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}^{<}_m(\mathbb{R})$, and whose open-to-closed part is empty. The algebra over this operad is the pair assembled from the boundary algebra and its centre; the bar coalgebra is the resolution by which the centre is computed:

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = R\mathrm{Hom}_{\mathcal{A}\otimes\mathcal{A}^{\mathrm{op}}}(\mathcal{A}, \mathcal{A}) = \mathrm{ChirHoch}^{\bullet}(\mathcal{A}, \mathcal{A}), \tag{15.1}$$

the chiral Hochschild complex, carrying cup product and brace operations extracted from insertion at boundary strata of FM_{k+1}, hence a homotopy Gerstenhaber (E_2) structure: the chiral form of Deligne's conjecture.

Theorem 15.1 (Terminality)**.** *The pair $\big(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\ \mathcal{A}\big)$, with the brace action of the centre on the algebra, is the terminal local open–closed pair over $\mathcal{A}$: every $\mathrm{SC}^{\mathrm{ch,top}}$-pair $(\mathcal{B}, \mathcal{A})$ admits a unique morphism of pairs $\mathcal{B} \to \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, given on elements by $b \mapsto (-1)^{|b|}\mu_{1;n}(b; -)$. Every local closed sector acting on a chiral algebra factors uniquely through its derived centre.*

Proof. The codimension-one identities of the Swiss-cheese operad force the assignment to be a chain map compatible with the braces; evaluation on the boundary algebra forces its value. Existence and uniqueness are the two halves of the same computation, and terminality follows. ☐

Theorem 15.1 is holography as a universal property. The third dimension is constructed: it is the normal collar that the centre construction manufactures. The theorem is also exactly as strong as such a statement can be:

Proposition 15.2 (Separation)**.** *A decoupled closed sector with zero boundary coupling acts trivially on every entry of the algebraic data*

$$\big(\mathcal{A},\ \mathcal{A}^{i},\ \mathcal{A}^{!},\ \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),\ r(z),\ \Theta_{\mathcal{A}}\big),$$

while enlarging the bulk observables of any holomorphic–topological theory that carries it. A physical bulk is therefore specified by the algebraic data together with an open–closed comparison map identifying $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ with its boundary observables: the comparison map is an independent coordinate of the theory.

15.2. **The annulus and the trace.** Evaluation on the circle produces the trace: by excision over the two-interval cover, the factorization homology of the annulus is the two-sided bar construction $\mathcal{A} \otimes^{L}_{\mathcal{A}\otimes\mathcal{A}^{\mathrm{op}}} \mathcal{A}$ — Hochschild homology — and the identification is Morita invariant. The cyclic structure of the trace is the datum that thickens graphs to ribbon graphs, and with it the genus expansion acquires its 't Hooft form: a matrix realisation of the algebra weights each stable graph by N^{χ}, and the planar limit of Section 11 is literal. The closed-sector trace, the cyclic symmetry of correlators, and the large-N counting are one structure, met three times.

15.3. **Topologisation.** The centre is holomorphic in the curve direction; it becomes topological exactly when translation is inner.

Theorem 15.3. *Let $\mathcal{A} = V_k(\mathfrak{g})$, $k \neq -h^\vee$, with Sugawara vector $T = \frac{1}{2(k+h^\vee)} \sum :J^a J_a:$. On the BRST cohomology of the derived centre, $T = [Q, G]$ for an explicit G; translations are therefore exact, the factorization algebra is locally constant on cohomology, and Dunn additivity upgrades the E_2-structure to E_3: the derived centre of an affine algebra at non-critical level is a three-dimensional topological object. At the critical level the Sugawara construction diverges and the topologisation collapses onto the commutative Feigin–Frenkel centre. For class M the statement holds on cohomology and on weight-completed models; on the original complex it is equivalent to an A_∞-coherent lift of $[m, G] = \partial_z$, the remaining step.*

Proof. $\partial_z \mathcal{O} = [T, \mathcal{O}] = [Q, [G, \mathcal{O}]]$ makes translation exact; a factorization algebra with exact translations is locally constant on cohomology; Lurie's recognition principle converts local constancy into a topological E_1-factor transverse to the holomorphic E_2; Dunn additivity multiplies them. □

15.4. **Three cohomologies.** Three cohomology theories shadow the centre, on three domains: the curve-level chiral Hochschild complex of this section; the Hochschild cohomology of a chosen mode algebra on the formal disc; and the Gelfand–Fuchs continuous cohomology of the positive modes. The comparison maps run from the curve to the disc — evaluate at a colliding configuration and set the spectral variables to zero — and from the disc to Gelfand–Fuchs by antisymmetrisation against an invariant functional; each map forgets the data proper to its source, the spectral variables in the first case and the full associative structure in the second. The three theories agree in low degrees on the benchmarks and diverge exactly where they should: the Gelfand–Fuchs side is unbounded where the chiral side is finite, and the difference is measured by the cone of the comparison, which is itself computable.

15.5. **Support of the centre.**

Theorem 15.4. *Suppose given a family datum for $\mathcal{A}$: a filtered model of* $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ *together with a strong deformation retract onto a complex supported in a finite set S of degrees. Then the support of* $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ *lies in S. The benchmarks, due to Bakalov–De Sole–Kac* [BDSK]*, are: for* Vir_c*, support $\{0, 2, 3\}$ with one-dimensional groups — the Poincaré polynomial is $1 + t^2 + t^3$, and cup product with the anomaly class is an isomorphism $H^0 \to H^2$; for the rank-one even superboson, support $\{0, 1\}$ with dimensions $(2, 1)$. When a perfect pairing of degree d aligns the two charts,* *the support is palindromic:* $\mathrm{ChirHoch}^n(\mathcal{A}) \cong \mathrm{ChirHoch}^{d-n}(\mathcal{A}^!)^\vee \otimes \omega_X$ *and* $P_{\mathcal{A}}(t) = t^d P_{\mathcal{A}^!}(t^{-1})$ *— at* $d = 2$ *the deformations of* $\mathcal{A}$ *are the obstructions of* $\mathcal{A}^!$*, exchanged by duality. For the affine, Virasoro and* $\mathcal{W}$ *families the retract from collision geometry is the hypothesis under which the theorem operates. At the critical level the support is infinite: the centre is the algebra of functions on opers, with Hilbert series* $\prod_{i=1}^r \prod_{m \geq 0}(1 - q^{d_i+1+m})^{-1}$*. Finiteness of the closed sector is a property of non-degenerate vacua, and it ends exactly where the Langlands-dual geometry begins.*

16. Arithmetic of the tower

16.1. **Two channels.** The shadow tower meets arithmetic through two channels of opposite transcendence, and the asymmetry between them is structural.

The *algebraic channel* is Theorem 6.1: shadow extraction is a depth-one residue against Arnold forms — no iterated integrals occur — so the tower of any single-line family is rational. For the Virasoro family more is true: at leading order in $1/c$ the recursion closes on a single plethystic primitive, and the entire tower is the iterated self-extension of one Kummer motive with parameter $-6/c$; the leading Laurent coefficients are $A_r = 8(-6)^{r-4}/r$, with the subleading tier $B_r = -A_r\left(\frac{22}{5} + \frac{(r-4)(r-5)}{18}\right)$ in closed form — the Lee–Yang constant $\frac{22}{5}$ governing the entire subleading stratum. Infinite algebraic depth, zero transcendence. The associator, with its multiple zeta values, lives in the kernel of the averaging map and never reaches the tower.

The *arithmetic channel* opens for lattice algebras, whose graph amplitudes factor through theta series in a finite-dimensional Hecke module. The tower then resolves the constrained Epstein zeta function one Hecke constituent per degree: degree two reads the constant term, degree three the Eisenstein part, degree $3 + j$ the j-th cusp eigenform, and termination is the dimension of the space of cusp forms. For the Leech lattice the quartic obstruction *is* the Ramanujan L-function $L(s, \Delta_{12})$. Finite depth, periods of full transcendence. The resolution is worked in rank: for $\mathbb{Z}$ the tower stops at once; for E_8 there is one Eisenstein and one cusp constituent; for the Leech lattice the theta series is $E_{12} - \frac{65520}{691}\Delta_{12}$, the Ramanujan congruence $\tau(n) \equiv \sigma_{11}(n) \bmod 691$ reading off the constant, and the tower terminates at depth four because $\dim S_{12}(\mathrm{SL}_2(\mathbb{Z})) = 1$.

16.2. **Irregular primes.**

Theorem 16.1. *Let* $Z_g(k)$ *be the genus-*g *Verlinde number of* $\widehat{\mathfrak{sl}}_2$ *at level* k*, a polynomial in* $n = k+2$ *of degree* $3g-3$*:* $Z_g = n^{g-1}\sum_{j=1}^{n-1} \csc^{2g-2}(\pi j/n)/2^{g-1}$*. Its leading coefficient is* $(-1)^g 2^{2g-1} B_{2g-2}/(2g-2)!$*, by the Hurwitz expansion of the cosecant against Euler's evaluation of* $\zeta(2g-2)$*. Consequently the Kummer-irregular primes enter the Verlinde polynomials through Bernoulli numerators:* 691 *divides the numerator of* $[n^{18}]Z_7$ *and* 3617 *divides that of* $[n^{24}]Z_9$*, each prime first appearing at genus* $g(p)$ *with* $p > 2g-2$*, so that the* *factorial factor is a p-adic unit. In the computed range $r \leq 11$ the cleared numerators of the Virasoro tower contain no Kummer-irregular prime: the two channels are disjoint where both have been computed.*

Proof. The cosecant power sum is polynomial in n by the Eisenstein–Cauchy argument; its two boundary tails give $2\zeta(2g-2)(n/\pi)^{2g-2}$ by Hurwitz's expansion, and Euler's evaluation converts $\zeta(2g-2)$ into B_{2g-2}. Witten's symplectic volume of the moduli of flat connections [Wit] provides the independent second derivation of the same leading coefficient. □

16.3. **The Galois layer.** The paramodular forms of the genus-two kernel carry Galois theory, and the first floor is computed. The spinor L-function of the Saito–Kurokawa lift attached to the Igusa form has Hecke eigenvalues $\lambda_p = a_p(f_{18}) + p^8 + p^9$ with $f_{18} = E_6\Delta$, its degree-four factor p^{34}; the associated deformation ring is non-reduced, $\mathbb{Z}_\ell[[x_1, x_2, x_3]]/(x_1^2, x_1x_2, x_1x_3)$, the nilpotent direction cut out by a cup-square in Galois cohomology — the arithmetic counterpart of a first-order deformation that obstructs itself, which is the Galois-theoretic face of the quartic obstructions this paper computes on the chiral side. The matching of the two obstruction theories across the Igusa kernel is the arithmetic form of the modular Koszul programme.

16.4. **Dirichlet lifts of the sewing operator.** The sewing spectrum organises into Dirichlet series, and the free and gravitational families lift differently:

$$\begin{gathered} S_{\mathcal{H}}(u) = \zeta(u)\,\zeta(u+1), \qquad S_{\mathrm{Vir}}(u) = \zeta(u+1)\big(\zeta(u)-1\big), \\ S_{\mathcal{W}_N}(u) = \zeta(u+1)\Big((N-1)\zeta(u) - \textstyle\sum_{j<N} H_j(u)\Big), \end{gathered} \tag{16.1}$$

the Heisenberg lift carrying the full Euler product of $\zeta(u)\zeta(u+1)$ — the logarithm of the Fredholm determinant of Section 13 expanded in divisor sums — and each $\mathcal{W}$-generator subtracting its own harmonic correction: the Miura transformation acts on the arithmetic side as a divisor-sum defect. For even unimodular lattices the depth of the Hecke resolution is $3 + \dim S_{r/2}$: depth two for $\mathbb{Z}$, three for E_8, four for the Leech lattice, whose cusp constituent is the Ramanujan form.

16.5. **The reach of genus one.** Three theorems fix the reach of the genus-one theory. Rankin–Selberg unfolding of any genus-one integral against an Eisenstein series retains exactly the constant term of the zeta function; the modular Maurer–Cartan constraint lives in the pluriharmonic direction, which Weil–Petersson geometry projects out; and the genus-one sewing operator is Fock-diagonal, so every two-variable Dirichlet series it produces collapses to one variable. Genus-one Maurer–Cartan data is integrability: it fixes the couplings. The location of zeros is a genus-two phenomenon: the sewing kernel of genus two carries an off-diagonal Schur complement, and through the Igusa cusp form it reaches the central values of twisted L-functions — the route the next paper takes.

17. Deformation quantization and the E_3 identification

17.1. **Quantization as Maurer–Cartan theory.** A coisson algebra — a commutative chiral algebra with a Lie* bracket — is quantized by lifting a Maurer–Cartan element order by order in $\hbar$ through its chiral deformation complex; the local input is formality on the disc, the global obstructions live in the modular directions, and the classification of quantizations is the gauge groupoid of the complex. Level quantization is the first example: for the affine algebra the obstruction lattice is $H^3(\mathfrak{g};\mathbb{Z})$, and integrality of the level is the integrality of the Maurer–Cartan class.

The boundary form of quantization is a theorem of Koszul type. For a potential F, the cofree coalgebra on the graph of dF carries a canonical coderivation, and the coderivation squares to zero exactly on the derived critical locus: for $F = x^2$ the square evaluates to $6c^2$ on the quartic cogenerator, and the terminal square-zero quotient is the Koszul complex of the equations. On that quotient the Hochschild cohomology is the algebra of functions on the shifted cotangent of the derived zero locus, $HH^\bullet(A_F) \simeq \mathcal{O}\big(T^*[-1]Z_F^{\mathrm{der}}\big) = \mathcal{O}\big(d\mathrm{Crit}\big)$ — Landau–Ginzburg theory as the simplest instance of the principle that curvature, unabsorbed, cuts out geometry.

17.2. **The identification with Chern–Simons theory.** For $\mathcal{A} = V_k(\mathfrak{g})$ at non-critical level the derived centre is small and completely computable on the formal disc:

$$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}\big(V_k(\mathfrak{sl}_2)\big) \;=\; \mathbb{C} \,\oplus\, \mathfrak{sl}_2[-1] \,\oplus\, \mathbb{C}[-2], \tag{17.1}$$

with vanishing cup product and Gerstenhaber bracket and with the P_3-bracket

$$\{X, Y\} \;=\; \frac{(X, Y)}{k + h^\vee} \tag{17.2}$$

as its only structure: the entire local closed sector of the Wess–Zumino–Witten model is the invariant form divided by the shifted level. On the other side, perturbative Chern–Simons theory for $\mathfrak{g}$ produces a filtered E_3-algebra of observables whose deformation space is $\lambda\, H^3(\mathfrak{g})[[\lambda]]$ with $\lambda = k + h^\vee$; Whitehead's lemma makes $H^3(\mathfrak{g})$ one-dimensional for simple $\mathfrak{g}$, formality of the E_3-operad converts both sides into P_3-deformation problems, and the two Maurer–Cartan classes agree because both are pinned by the binary bracket above. The identification of the two formal families is the precise statement that the derived centre of the affine algebra is the observable algebra of quantum Chern–Simons theory on the formal disc; the representative depends on the associator, through the same torsor as everything else in Section 4.

17.3. **The operadic circle.** The tower of structures closes into a circle:

$$E_3 \;\longrightarrow\; E_2 \;\longrightarrow\; E_1 \;\longrightarrow\; E_2 \;\longrightarrow\; E_3, \tag{17.3}$$

the topological bulk determining the braided category of its lines, the braided category its chiral boundary algebras, the boundary algebra its ordered bar complex, the bar complex the Drinfeld double, and the double — through its Drinfeld centre — the bulk again. For the Heisenberg algebra the circle closes as a theorem: the Drinfeld centre of the completed double-module category and the derived chiral centre are both three-dimensional with trivial P_3-structure, identified by module–comodule duality composed with Hochschild–coHochschild duality. On the whole Koszul locus the Morita comparison closes; the closing of the circle for the gravitational classes is the categorical form of holography, and it is the natural endpoint of the programme of Section 15.

18. The Calabi–Yau source

18.1. **The Atiyah–Connes class.** A Calabi–Yau category carries a canonical obstruction to chiral specialisation, and it is a characteristic class. Let X be a smooth projective Calabi–Yau threefold with its cyclic dg-enhancement: the Connes operator Δ_2 transports the cyclic differential to the cochain level, the ternary product m_3 realises the pentagonal cell of the Deligne structure, and their commutator is a closed four-cochain whose Hochschild–Kostant–Rosenberg projection, contracted with the holomorphic volume form, is

$$\alpha_X \;=\; \iota_{\Omega_X}\, \pi^{2,2}_{\mathrm{HKR}}\big[m_3, \Delta_2\big] \;\in\; H^{1,2}(X), \tag{18.1}$$

represented in Dolbeault cohomology by $\iota_{\Omega_X}\,\mathrm{tr}(\bar\partial\nabla \wedge \bar\partial\nabla)$ for any connection, independently of the choice: the Atiyah class of X, curled by the cyclic structure.

Theorem 18.1 (Spine)**.** *The two-stage specialisation — integrate over a surface, restrict to a curve — carries the Atiyah–Connes class to the first bar–cobar obstruction of the resulting chiral algebra: $sp_{\Sigma,C}(\alpha_X) = o_3^{(0)}(A_C)$. In particular the vanishing of α_X forces the vanishing of the first obstruction of every chiral algebra the threefold produces.*

Proof. The two-stage specialisation is a filtered morphism of cyclic deformation complexes, and both sides of the equation are its degree-two shadow: integration over Σ carries the Hochschild–Kostant–Rosenberg $(2,2)$-projection to the binary collision residue, restriction to C evaluates it, and the class $[m_3, \Delta_2]$ maps to the first obstruction cocycle by naturality of the bar differential under both operations. On $S \times E$ the evaluation is one pairing: both sides land on the line $H^2(S, \mathcal{O}_S) \otimes H^0(E, \Omega^1_E)$, and the morphism identifies the two coefficients. □

The spine is the bridge between the geometric source and the chiral theory of this paper: an obstruction living on a threefold, measured by a Hodge class, becomes an obstruction living on a curve, measured by the tower of Section 6. On a product $S \times E$ of a $K3$ surface and an elliptic curve the class concentrates, by Künneth naturality, on the one-dimensional line $H^2(S, \mathcal{O}_S) \otimes H^0(E, \Omega^1_E)$: the entire question of specialisation is one scalar, and its expected value is proportional to $\int_S c_2 = 24$.

18.2. **The chirality ladder.** The chiral output of a Calabi–Yau category is graded by dimension: $d = 1$ produces E_∞-chiral algebras, $d = 2$ produces E_2-structures through the Drinfeld centre, and $d \geq 3$ stabilises at E_1 — the ordered theory of Section 4 is the generic output of the geometric source, which is the deepest reason the ordered theory is primary. The scalar shadow stratifies with the Hodge theory: at odd d the Serre pairing cancels the supertrace in pairs and the chiral scalar vanishes; at even d the middle Hodge number survives, giving $\kappa_{\mathrm{ch}}(K3) = 2$ and its analogues. The cohomological Hall algebra of the resolved geometry supplies the positive half of the quantum group, its Drinfeld double the whole; for the $K3$ surface the resulting object has twenty-four generators indexed by the Mukai lattice, with the abelian Drinfeld coproduct $\Delta_z x_i^\pm(u) = x_i^\pm(u) \otimes 1 + 1 \otimes x_i^\pm(u - z)$, Serre relations governed by the 24×24 intersection matrix, and operator products computed to the order the Serre relations require — the quantum group of a surface, with the surface's arithmetic in its Cartan matrix.

18.3. **One quantum group.** The families of Section 4 and Section 8 are specialisations of a single object $Q_{\mathfrak{g}}^{k,f,\mu}$, parametrised by a Lie type, a level, a nilpotent, and a spectral parameter: the Yangian at $k = 0$, $f = 0$; the affine Yangian at generic level; the shifted Yangians by change of oscillator variables; the Coulomb-branch algebras of three-dimensional gauge theories; and the $\mathcal{W}$-algebras, principal and not, through Drinfeld–Sokolov reduction. One object, eight families, one duality: the inversion principle of Theorem 4.2 acts on $Q_{\mathfrak{g}}^{k,f,\mu}$ and restricts to every specialisation.

19. Lattices, moonshine, and the Borcherds boundary

19.1. **Self-duality and the census.** For a lattice algebra V_Λ with 2-cocycle ε, the inversion principle of Theorem 4.2 gives $(V_\Lambda^\varepsilon)^i = (V_\Lambda^{\varepsilon^{-1}})^c$; for even unimodular Λ with symmetric cocycle the bar coalgebra is self-dual. The lattice algebras are also the one place where a genuinely E_1 chiral algebra arises classically: a non-symmetric cocycle produces a braiding that is an independent input, not derived from a vertex algebra. The holomorphic $c = 24$ census organises itself by the weight-one threshold: the Schellekens list [Sch] decomposes as $71 = 24 + 1 + 46$ — twenty-four Niemeier lattice algebras, the Moonshine module as the unique entry whose weight-one space vanishes, and forty-six orbifolds — and the classification data of Section 6 refine the census: all lattice entries are class G with $\kappa = 24$, the Moonshine module is class M with $\kappa = 12$.

19.2. **The Borcherds boundary.** The strictification mechanism of Section 4 — the obstruction to a strict quantum-group structure lives in root spaces and dies when all root multiplicities are one — fails exactly for Borcherds algebras, whose imaginary simple roots have multiplicity. This is the method locating the automorphic boundary of the theory. On that boundary sit the denominator identities: the Igusa cusp form factors as $\Phi_{10} = \Delta_5^2$, with Δ_5

the Borcherds product of weight $5 = c_{\Delta_5}(0)/2$ attached to the $K3$ elliptic genus by Gritsenko–Nikulin [GN], and the second-quantised $K3$ elliptic genus of Dijkgraaf–Moore–Verlinde–Verlinde [DMVV] is the inverse of the Igusa form,

$$\sum_{N \geq 0} p^N \chi_{\mathrm{ell}}\big(\mathrm{Sym}^N K3\big) = \frac{p\,q\,y}{\Phi_{10}(p,q,y)}, \tag{19.1}$$

the symmetric-orbifold linearity of Section 7 exponentiated into a Borcherds product. The weight formula $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(0)/2$ holds across the paramodular family, with the weights $5, 3, 2, \frac{3}{2}, 1$ at $N = 1, 2, 3, 4, 6$: one arithmetic law, five instances, all Borcherds products. Against this proved automorphic arithmetic stands Conjecture 10.4: that the chiral algebra recognised from the $K3$ source carries conductor $2c_+ = 8$ and quantisation $\hbar^2 = -1/8$. The five scalars attached to $K3 \times E$ — categorical 0, Hodge 0, Heisenberg 3, Borcherds weight 5, fibre Euler number 24 — are five distinct invariants, and the conjecture identifies exactly one of them with the conductor. An order-two datum makes the distinction vivid: the physical partition function is Δ_5^{-2}, and the orientation bit that distinguishes the two candidate chiral objects is remembered by the algebra and integrated away by its partition function — the sharpest instance of the principle that the object carries strictly more than its scalar shadow.

19.3. **What the tower sees of moonshine.** The shadow tower sees the Monster through its coefficients: for $V^\natural$, $\Delta = 20/71$ and the tower is asymptotic with the same Borel structure as Ising; the McKay–Thompson refinements enter as twining fingerprints; and moonshine enters the classification as boundary data, the automorphic side of the correspondence standing on its own theorems.

20. The physics

Each statement of this section is a theorem of the formalism; where a comparison map with a physical theory enters, it enters in the statement.

20.1. The anomaly is the curvature.

Theorem 20.1 (Rigidity of the anomaly)**.** *The obstruction to a translation-invariant chiral coproduct on Vir_c is the class $(c/2)\,\Omega^{\mathrm{univ}} \in H^2_{\mathrm{ch}}(\mathrm{Vir}_c, \mathrm{Vir}_c)$: a z-independent coproduct exists if and only if $c = 0$, and the coefficient $c/2 = \kappa(\mathrm{Vir}_c)$ is stable under pullback along modules. The conformal anomaly of two-dimensional physics, the fourth-order pole of the stress tensor, and the modular curvature of Theorem 5.1 are one number.*

The genus-one free energy is triply determined: as the degree-two projection of $\Theta_{\mathcal{A}}$, as the logarithmic derivative $\partial_\tau \log \eta^{-2\kappa}$, and as the regularised Casimir energy on the circle — three computations, one number, $F_1 = \kappa/24$. The string-theoretic identities follow at scalar level. The ghost system carries $\kappa(bc_{\lambda=2}) = -13$; matter plus ghosts is curvature-free precisely when $c = 26$, which is the identity $\varkappa(\mathrm{Vir}) = 13$ of Theorem 10.1 read as anomaly cancellation, with the self-dual point $c = 13$ at its centre. The genus-zero bar complex of the $c = 26$ system is identified with the BRST complex under a filtered comparison; nilpotence of the BRST charge, $Q^2 = \frac{c-26}{12}\sum_n (n^3 - n) :c_{-n}c_n:$, is the uncurved condition.

20.2. **The two central charges.** The theory distinguishes two exceptional values of c and assigns each its own mechanism. At $c = 26$ the *dual* curvature vanishes: $\kappa(\mathrm{Vir}_{26-c}) = 0$, matter and ghosts cancel, and the total bar differential of the coupled system is uncurved — the critical dimension is the flatness locus of the Koszul-dual theory. At $c = 13$ the theory is *self-dual*: the fixed point of the involution, the balance point of the entropy sum rule, the centre of the conductor. Both values are visible already in (10.3), as the zero and the fixed point of one involution; the string selects the first, the symmetry the second.

20.3. **The anomaly ratio.** The ratio $\varrho = \kappa/c = \sum_i 1/(m_i + 1)$, summed over the exponents of the underlying Lie algebra, is a spectral invariant of the family: 1 for the Heisenberg lane, $\frac{1}{2}$ for Virasoro, $\frac{5}{6}$ for $\mathcal{W}_3$, $\frac{13}{12}$ for $\mathcal{W}_4$, $\frac{121}{126}$ for the E_8 family. It converts between the two conductors, $\varkappa = \varrho\, K$, and it measures what survives Drinfeld–Sokolov reduction: the exponents are the surviving weights, and the ratio is their harmonic signature. That a single rational number carries the exponent spectrum of the reduction into the modular curvature is the kind of bookkeeping identity the subject produces when the definitions are right.

20.4. **The string measure.** For the rank-r Heisenberg algebra the analytic theory closes. The sewing operator on the weighted Fock space is trace class with eigenvalues q^n; its Fredholm determinant is $\det(1 - T_q) = q^{-1/24}\eta(q)$; and the genus-$g$ partition function, assembled by sewing, is

$$Z_g \;=\; \big(\det \operatorname{Im}\Omega\big)^{-r/2}\big(\det\nolimits'_\zeta \Delta_{\Sigma_g}\big)^{-r/2}, \tag{20.1}$$

the Polyakov measure of the bosonic string [DP], derived here by operator theory from the bar complex, with the Belavin–Knizhnik form as cross-check. Lattice algebras multiply this by the Siegel theta series; at genus two the theta series separates $E_8 \oplus E_8$ from D_{16}^+, a separation genus two alone achieves.

20.5. **Soft limits and primitivity.** The first shadows have names in the flat-space expansion of gravity: $S_2 = c/2$ is the leading (Weinberg) soft coefficient, $S_3 = 2$ the subleading (Cachazo–Strominger) coefficient, and $S_4 = 10/[c(5c + 22)]$ the first coefficient beyond, entering through the Mellin residue at $\Delta_s = 4 - r$; the higher soft expansion proceeds through the Mellin residues, each shadow coefficient feeding the pole at its own depth. A cleaner statement is coalgebraic: for every class-M algebra obtained by Drinfeld–Sokolov reduction, the coproduct on the dual is strictly primitive — a ghost-number obstruction kills all group-like corrections — while the A_∞ tower is infinite. Gauge theory deforms the coproduct and truncates the tower; gravity does the opposite. Gravity braids particles; braiding is its entire product.

20.6. **Three-dimensional gravity.** The Virasoro instance of Theorem 15.1 packages the boundary algebra at Brown–Henneaux central charge $c = 3\ell/2G$, the bulk $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$ with Poincaré polynomial $1 + t^2 + t^3$, and the collision kernel $r_c(z) = (c/2)z^{-3}+2Tz^{-1}$ into an algebraic holographic sector for three-dimensional gravity. Within it: BTZ states are Maurer–Cartan elements; the Cardy entropy is the character asymptotics under modular invariance and vacuum dominance; and the two-branch entropy shadow crosses at $t_* = 3S_{\mathrm{BH}}/13$, the coefficient being c-independent because the rate sum is $\varkappa/3 = 13/3$. In the Cardy regime the entropy is character asymptotics: given modular invariance of the torus block and dominance of the vacuum channel, the density of states at large weight gives $S = 2\pi\sqrt{c\Delta/6}+2\pi\sqrt{c\bar{\Delta}/6}$, with the determinant of the modular transformation supplying the $-\frac{3}{2}\log S$ correction; for the Brown–Henneaux value of c this is the horizon area over $4G$. The crossing formula is a theorem; the two-branch structure realises Conjecture 10.3, whose scalar shadow $\varkappa = 13$ it uses; and the dynamical-metric completion of the sector is the frontier of the subject, with the sum over topologies replaced by the Maurer–Cartan induction of Theorem 5.5 — an induction, every term finite.

20.7. **Chern–Simons theory and knots.** The abelian case anchors the dictionary at the level of theorems: the Heisenberg algebra $\mathcal{H}_k$ is the boundary algebra of abelian Chern–Simons theory at level k, the boundary operator product $k/(z - w)^2$ arising from the slab Green's function, and $\kappa = k$ identifies the modular curvature with the Chern–Simons level; the Wilson-line framing phase $e^{2\pi i q^2/k}$ is the boundary value of the R-matrix $e^{k\hbar/z}$. In the nonabelian case the ordered bar complex of $\widehat{\mathfrak{g}}_k$ carries the braid monodromy of the quantum group at $q = e^{\pi i/(k+h^\vee)}$; closure of braids through the genus-one trace, together with the standard ribbon-category trace, computes the coloured Jones polynomials, the writhe correction being the Casimir framing anomaly. Level–rank and boundary dualities are instances of Koszul duality of boundary conditions; the identification of the Yangian with the line-operator algebra of four-dimensional Chern–Simons theory [CWY] is the physical face of Theorem 4.3, with the non-renormalisation of the line OPE as the physical face of Koszulness.

20.8. **Wilson lines and evaluation modules.** Line operators of the three-dimensional theory are evaluation modules of the ordered dual: the spectral parameter is the holomorphic position of the line, braiding of lines is the R-matrix, and the operator product of parallel lines is one-loop exact — the non-renormalisation theorem of the physics is the collapse of the bar spectral sequence, which is Koszulness. Monopole operators arrive with the affine Grassmannian, and the Coulomb branches of three-dimensional gauge theories realise the shifted Yangians [BFN]: the ordered theory of

Section 4 is the algebra of the line-operator sector, stated once and specialised everywhere.

20.9. **Entanglement.** The genus-one shadow assigns to an interval the entropy $S_{EE} = (c/3)\log(L/\varepsilon)$, and complementarity of Theorem 10.1 gives the sum rule $S_{EE}(\mathrm{Vir}_c) + S_{EE}(\mathrm{Vir}_{26-c}) = \frac{26}{3}\log(L/\varepsilon)$, saturated symmetrically at $c = 13$. The classification grades entanglement complexity: Gaussian for class G, a single logarithm for L, a quartic contact correction for C, an infinite Rényi tower for M. The Lagrangian splitting of Theorem 10.1 is a symplectic code — each summand isotropic, the cross-pairing perfect — and a Hilbert-space datum completes it to a quantum code; the formulas of this subsection are shadow functionals, exact as stated.

20.10. **Integrable structures.** The genus-zero shadow connection of the affine algebra is the Knizhnik–Zamolodchikov connection with coupling $1/(k + h^\vee)$; flatness is the Arnold relation contracted with the Casimir, which is the classical Yang–Baxter equation; the Gaudin Hamiltonians are its residues. At genus one the propagator degenerates through the Weierstrass zeta function to the elliptic r-matrices of Belavin and the dynamical connection of Knizhnik–Zamolodchikov–Bernard and Felder [Fel], and the B-cycle monodromy exponentiates the coupling to $q = e^{\pi i/(k+h^\vee)}$; braid monodromy of the ordered bar complex computes the coloured Jones polynomials after the standard ribbon-category trace. All three appearances of $\hbar$ — the KZ coupling, the line-operator loop parameter, the bar coupling — are the same number $1/(k + h^\vee)$, and this identification is forced by the Sugawara construction.

21. The one question

Every construction in this paper is a projection of the single element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$: the r-matrix and the Knizhnik–Zamolodchikov connection at genus zero; κ and the curvature at degree two; the shadow tower along a line; the complementarity splitting and its conductor over $\overline{\mathcal{M}}_g$; the genus expansion as a trace; the Verlinde numbers on the integrable quotient; the quantum group in the ordered kernel. Everything proved is a finite-order projection of $\Theta_{\mathcal{A}}$. Everything open is a question about its ambient.

Question 21.1. Characterise the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ as a geometric object.

The evidence assembled here proposes an answer: $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ is the endomorphism algebra of a Lagrangian embedding $\mathcal{L}_{\mathcal{A}} \hookrightarrow \mathcal{M}_{\mathrm{vac}}(\mathcal{A})$ in a (-2)-shifted symplectic stack of vacua. The shadow tower computes its Taylor jets; the Swiss-cheese structure of Section 15 its local composition law; the shift matches the $-(3g-3)$-shifted symplectic structure of Theorem 10.1. The remaining object is the global geometry: the stack itself, the moduli of its Lagrangians, and the descent that makes the construction independent of all choices. The known boundary of the theory — the critical level, where $\kappa = 0$

and the centre becomes the algebra of functions on opers; the logarithmic algebras outside the classification; the multi-weight corrections that leave $\mathbb{Q}(c)$; the Borcherds algebras where the strictification mechanism fails and the automorphic forms begin — marks exactly where that geometry must be nontrivial.

Five problems structure the frontier, and each is a construction.

Chain-level topologisation. Lift the E_3-structure of Theorem 15.3 from cohomology to the original complex for the gravitational classes: the coherent tower over $[m, G] = \partial_z$, expected to close in the coderived ambient, where the obstruction cocycles are curvature-torsion.

Nodal descent. Extend Theorems 3.2 and 10.1 across the boundary of $\overline{\mathcal{M}}_{g,n}$ with prescribed clutching: logarithmic sewing data at the separating and non-separating divisors, completing the modular functor at chain level.

The multi-weight generating function. A closed form for the cross-channel corrections of Theorem 11.2 in the Frobenius geometry of the $\mathcal{W}_N$ families, where the computed values at genus two, three and four already fix the shape.

Compact completion. Extend the ordered duality of Theorem 4.3 from the evaluation core to the full module category: in type A a single compact-generation statement on the Baxter locus.

The dynamical metric. Complete the algebraic holographic sector of Section 20 to a gravitational path integral, with the Maurer–Cartan induction as the sum over topologies and the Igusa kernel as the genus-two input.

One 1-form, one relation, one extraction, one ordering.

Appendix A. The landscape census

algebra	*class*	κ	*dual partner*	$\varkappa$
$\mathcal{H}_k$	G	k	curved Sym	0
free fermion	G	$1/4$	itself	—
V_Λ, Λ unimodular	G	$\operatorname{rk}\Lambda$	itself (cocycle inverted)	0
$V_k(\mathfrak{g})$, $k \neq -h^\vee$	L	$\frac{(k+h^\vee)\dim\mathfrak{g}}{2h^\vee}$	level $-k-2h^\vee$	0
$\widehat{\mathfrak{g}}_1$, simply laced	G	$\operatorname{rk}\mathfrak{g}$	lattice realisation	0
$\beta\gamma_\lambda$	C	$6\lambda^2 - 6\lambda + 1$	bc_λ	0
Vir_c	M	$c/2$	Vir_{26-c} (Conj. 10.3)	13
$\mathcal{W}_N$	M	$c(H_N - 1)$	level-reflected $\mathcal{W}_N$	$(H_N-1)K_N$
$\mathcal{W}^k(\mathfrak{sl}_3, f_{\min})$	mixed	see §8	$k \mapsto -k-6$	c-conductor 50
$V^\natural$	M	12	Virasoro sector partner	—
Leech V_Λ	G	24	itself	0
$Y_\hbar(\mathfrak{sl}_2)$	E_1	—	$Y_{-\hbar}(\mathfrak{sl}_2)$	—
$V_{-h^\vee}(\mathfrak{g})$	FF	0	$\mathrm{Fun}\,\mathrm{Op}_{^L\mathfrak{g}}(D)$	0

The classes are those of Theorem 6.3; the $\mathcal{W}_N$ conductor carries the Drinfeld–Sokolov transport hypothesis of Theorem 10.2; the central-charge conductors $c + c^!$ are 26, 100, $4N^3 - 2N - 2$, $2\dim\mathfrak{g}$, 50 on their respective rows, with $K_N = 4N^3 - 2N - 2$, exact as rational functions of the parameter.

Appendix B. Notation

$\eta_{ij} = d\log(z_i - z_j)$	the logarithmic propagator
$\mathrm{FM}_n(X)$	Fulton–MacPherson compactification of n points
$\overline{B}^{\mathrm{ord}}_X(\mathcal{A})$, $\overline{B}^{\Sigma}_X(\mathcal{A})$	ordered and symmetric bar coalgebras
$\mathcal{A}^i$, $\mathcal{A}^!$	quadratic dual coalgebra; Koszul dual algebra
$K_X(\mathcal{A}) = \mathbb{D}_{\mathrm{Ran}}\overline{B}_X(\mathcal{A})$	Verdier algebra of the bar coalgebra
$\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$	derived chiral centre (chiral Hochschild complex)
$\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$	the modular Maurer–Cartan element
$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$	modular convolution Lie algebra
$\kappa(\mathcal{A})$	modular characteristic: scalar of the genus curvature
S_r, $H(t)$, $Q(t)$	shadow tower, its generating function, its square
$\Delta = 8\kappa S_4$	critical discriminant of the tower
$r_{\max}$	shadow depth; classes $\mathsf{G}, \mathsf{L}, \mathsf{C}, \mathsf{M}, \mathsf{FF}$
$\varkappa = \kappa + \kappa^!$	conductor; $K = c + c^!$ its central-charge form
λ_g^{FP}	Faber–Pandharipande Hodge numbers
$r(z)$, $R(z)$	classical and quantum r-matrices
$E(z, w)$	the prime form; $\omega_g = c_1(\mathbb{E}_g)$ the Hodge class
$q = e^{\pi i/(k+h^\vee)}$	the quantum parameter of level k

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